

TECHNICAL PROPOSAL

V1.2

**FOR THE DESIGN, DEVELOPMENT, IMPLEMENTATION,
TRAINING AND SUPPORT OF A HIGHER EDUCATION
ENTERPRISE RESOURCE PLANNING PLATFORM FOR TICH
IN AFRICA**

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Date:
2 July, 2026

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1.1: INTRODUCTION

The TICH Enterprise Resource Planning (ERP) Platform is a comprehensive digital transformation platform designed to integrate and automate the core academic, administrative, financial, and operational functions of TICH into a single centralized information system. The platform aims to eliminate fragmented processes, improve operational efficiency, enhance service delivery, strengthen institutional governance, and support evidence-based decision-making through real-time access to reliable information.

Following extensive stakeholder consultations, departmental interviews, workflow observations, document reviews, and operational audits, it became evident that many institutional processes are performed manually or through disconnected systems. Critical activities such as student admissions, academic administration, financial management, human resource operations, procurement, quality assurance, research management, and communication rely heavily on paper-based procedures, spreadsheets, and isolated databases. These approaches have resulted in duplicated data, delays in approvals, inconsistencies in reporting, limited transparency, and increased operational costs.

The proposed ERP platform seeks to address these challenges by establishing a unified digital ecosystem where all institutional departments operate on a shared database while maintaining appropriate security controls and role-based access. Information entered once into the system becomes available, subject to authorization, across all relevant departments, thereby eliminating repetitive data entry and improving institutional coordination.

Unlike conventional student management systems that focus primarily on admissions and examination processing, the proposed solution has been designed as a full Enterprise Resource Planning System. It supports the complete institutional lifecycle, beginning with prospective student recruitment and admissions, continuing through academic delivery, financial management, quality assurance, research, human resource administration, procurement, alumni engagement, and extending to community outreach programs and post-graduation services.

The platform has also been designed to support TICH's unique operational model, which includes centralized administration at the main campus, decentralized community colleges operating across multiple sub-counties, competency-based education and training (CBET), nursing block training, work-study programs, SACCO services, donor-funded projects, and regulatory compliance with Kenyan higher education authorities.

To maximize accessibility and operational efficiency, the ERP will be delivered through multiple digital channels. These include a public institutional website, secure web-based management portals, dedicated student and staff portals, a mobile application developed using Flutter, a desktop application and RESTful APIs that facilitate secure communication between the various system components. The platform will additionally incorporate automated notifications through email and SMS, AI-powered virtual assistance, secure payment gateway integration, and advanced analytics dashboards for institutional leadership.

Security, scalability, maintainability, and regulatory compliance constitute the architectural foundation of the proposed solution. The platform will implement Role-Based Access Control (RBAC), Multi-Factor Authentication (MFA), encrypted communications, comprehensive audit logging, automated backups, disaster recovery mechanisms, and compliance with the Kenya Data Protection Act, internationally recognized information security principles, and institutional governance policies.

Ultimately, the implementation of the TICH ERP Platform represents an institutional transformation program that modernizes business processes, enhances accountability, strengthens collaboration across departments, improves service delivery for students and staff, and establishes a sustainable digital foundation capable of supporting the institution's long-term strategic growth.

1.1 Background of TICH

TICH is a higher education institution committed to delivering quality education, vocational training, community-based development programs, research, and professional capacity building. Over the years, the institution has expanded its academic offerings, administrative functions, partnerships, and community outreach initiatives, resulting in increasingly complex operational processes that require efficient coordination across multiple departments.

The institution currently comprises several functional divisions, including Administration, Academics, Finance, Human Resources, Quality Assurance, Procurement, Research, Student Affairs, Information and Communication Technology (ICT), Community Colleges, SACCO operations, and executive management. Each department performs specialized responsibilities while relying on information generated by other departments to accomplish institutional objectives.

As student enrollment continues to increase & expected to increase and institutional programs diversify, existing manual processes have become increasingly difficult to manage. Many operational activities involve repetitive paperwork, multiple approval stages, manual reconciliation of financial transactions, spreadsheet-based reporting, and duplicated data entry across departments. These practices increase the likelihood of human error, reduce operational efficiency, and make it difficult for management to obtain timely and accurate information for strategic decision-making.

A comprehensive requirement gathering exercise involving interviews, departmental workshops, workflow analysis, and operational observations was conducted to understand the institution's existing business processes, operational challenges, approval hierarchies, regulatory obligations, and long-term strategic goals. The findings confirmed the need for an integrated enterprise platform capable of managing all institutional operations through a centralized and secure digital environment.

The proposed ERP platform has therefore been designed specifically around TICH's verified operational workflows rather than adopting generic higher education software. Every functional module, approval process, business rule, and user role has been modelled based on actual institutional practices observed during the requirement gathering exercise. This ensures that the final system aligns with existing operational realities while introducing automation, standardization, accountability, and opportunities for future institutional growth.

By implementing the TICH ERP Platform, the institution will establish a unified digital infrastructure that supports efficient service delivery, promotes transparency, strengthens regulatory compliance, improves data quality, enhances collaboration among departments, and provides management with real-time insights necessary for effective planning and decision-making.

1.2 Problem Statement

The rapid growth of TICH in academic programs, decentralized community colleges, research initiatives, and institutional support services has significantly increased the complexity of managing daily operations. While the institution has continued to expand its educational and administrative activities, many of its core business processes remain largely dependent on manual procedures,

paper-based documentation, spreadsheets, and isolated software tools such as quickbook and workpay that operate independently of one another.

The comprehensive requirement-gathering exercise conducted across the institution—including interviews with senior management, departmental heads, administrative personnel, finance officers, lecturers, quality assurance personnel, ICT staff, and students—revealed several operational inefficiencies that negatively affect institutional performance, service delivery, decision-making, and regulatory compliance.

One of the most significant challenges identified is the **fragmentation of institutional data**. Student records, staff information, financial transactions, academic records, procurement activities, quality assurance documents, and administrative reports are maintained in different locations using separate files and manual registers. As a result, departments frequently duplicate information, spend considerable time verifying records, and experience inconsistencies when generating institutional reports.

The admissions process similarly relies on multiple manual verification stages involving several departments. Applicant information is reviewed independently by academic staff, administration, and finance before admission decisions are finalized. Payment verification often requires manual confirmation, while admission letters and registration numbers are generated through separate processes. These disconnected procedures increase processing time, create opportunities for human error, and reduce the overall efficiency of student onboarding.

Academic administration presents additional operational challenges. Curriculum planning, course creation, unit allocation, timetable preparation, lesson planning, examination moderation, and student progression management require extensive coordination among the Academic Registrar, Heads of Department, tutors, and the Academic Board. Under the current operational model, much of this coordination depends on manual communication, spreadsheets, and printed documentation, making it difficult to maintain consistency, manage curriculum changes efficiently, and prevent scheduling conflicts involving lecturers, classrooms, and students.

Financial management is similarly affected by fragmented workflows. Student fee structures, invoice generation, payment verification, financial clearance, procurement approvals, payroll processing, statutory deductions, budgeting, and financial reporting are managed through multiple independent processes that require frequent manual reconciliation. Delays in payment verification directly affect course registration, examination eligibility, graduation clearance, and institutional reporting. Furthermore, the absence of real-time integration between financial transactions and student records limits the institution's ability to provide immediate financial visibility to both management and students.

Human Resource operations also rely heavily on manual administration. Employee onboarding, contract management, leave administration, payroll preparation, statutory compliance, staff performance monitoring, and professional development records are maintained through independent processes that require repeated data entry and manual tracking. This increases administrative workload while limiting management's ability to monitor workforce performance and contractual obligations efficiently.

The institution's decentralized community college model introduces additional operational complexity. Training centres distributed across multiple sub-counties require synchronized

admissions, student registration, attendance tracking, academic monitoring, and communication with the central campus. Without an integrated digital platform, maintaining consistent academic standards and operational visibility across geographically distributed centers becomes increasingly difficult.

Quality Assurance and Procurement departments experience similar challenges. Internal quality audits, compliance monitoring, supplier verification, budget approvals, procurement workflows, and institutional policy reviews are largely coordinated through manual documentation and email communication. These approaches reduce transparency, complicate audit preparation, and make it difficult to monitor the status of institutional compliance activities in real time.

Communication across the institution is another significant concern. Students, lecturers, administrators, and management often rely on separate communication channels to receive announcements, timetable updates, fee reminders, examination notifications, and administrative directives. The absence of a centralized communication platform increases the risk of delayed information dissemination and inconsistent messaging.

Institutional reporting also presents considerable challenges. Executive management requires timely access to accurate information regarding student enrolment, academic performance, financial status, staff performance, procurement activities, research projects, and quality assurance indicators. However, because operational data is distributed across multiple systems and manual records, report generation is time-consuming and often requires extensive manual compilation before meaningful insights can be obtained.

Security and regulatory compliance represent another area requiring improvement. The institution manages highly sensitive student, staff, financial, medical, and academic information that must be protected against unauthorized access, accidental modification, and data loss. Existing manual processes provide limited mechanisms for audit trails, role-based access control, data encryption, and comprehensive activity monitoring, increasing institutional risk while making compliance with the Kenya Data Protection Act and other regulatory requirements more challenging.

Collectively, these operational challenges reduce institutional efficiency, increase administrative costs, delay service delivery, and limit management's ability to make informed strategic decisions. They also create unnecessary workloads for staff while affecting the overall experience of students and external stakeholders.

The implementation of an integrated Enterprise Resource Planning (ERP) Platform is therefore necessary to consolidate institutional operations within a single secure information system. The proposed solution will automate critical workflows, centralize institutional data, standardize operational procedures, strengthen security, improve communication, enhance regulatory compliance, and provide real-time visibility into all major institutional activities. By addressing the identified challenges through a unified digital platform, TICH will establish a scalable technological foundation capable of supporting its continued institutional growth while improving efficiency, accountability, and service delivery across all departments.

1.3 Summary of Key Challenges Identified

The requirement-gathering exercise identified the following major institutional challenges that the proposed ERP platform is designed to address:

- **Fragmentation of Institutional Data:** Student records, financial transactions, staff data, and procurement activities are currently managed in isolated databases, spreadsheets, and manual registers. This fragmentation leads to possible duplicated data, increased time spent verifying records, and can lead significant inconsistencies when generating institutional reports.
- **Inefficient, Manual Operational Workflows:** Core processes such as student admissions, academic administration, and financial management mostly rely on paper-based documentation and manual approvals. These manual touchpoints delay service delivery, create opportunities for human error, and increase the administrative workload for staff.
- **Limited Management Visibility:** Because operational data is not centralised, executive management lacks real-time insights into institutional performance, such as enrollment trends, financial status, or procurement activities. Report generation is currently a time-consuming, compilation process.
- **Inadequate Security and Regulatory Compliance:** Existing processes offer limited mechanisms for audit trails, role-based access control, and data encryption. This increases institutional risk and makes maintaining compliance with the Kenya Data Protection Act and other regulatory requirements particularly challenging.
- **Communication Silos:** The lack of a centralized platform forces students and staff to rely on disjointed channels for critical updates, such as timetable changes, fee reminders, and administrative directives, leading to delayed or inconsistent information dissemination.
- **Financial Reconciliation Delays:** Financial management is hindered by multiple independent processes for invoicing, payment verification, and fee clearance, which require frequent and time-consuming manual reconciliation.
- **Academic Coordination Hurdles:** The coordination of curriculum planning, unit allocation, and timetable preparation relies on manual communication and spreadsheets, which may make it difficult to prevent scheduling conflicts for students and lecturers.
- **Fragmented Student Admissions:** The admissions process involves multiple manual verification stages across different departments, which may create bottlenecks and increases the time required for student onboarding.
- **Operational Transparency Gaps:** Many quality assurance, procurement, and policy review activities are coordinated through informal channels like email, which reduces overall transparency and complicates preparation for institutional audits.
- **HR Administration Burdens:** Human Resource tasks such as employee onboarding, contract management, and performance monitoring are maintained through independent, repetitive processes that limit management's ability to monitor workforce performance effectively.

- **Delayed Financial Visibility:** The lack of real-time integration between financial transactions and student records prevents management and students from accessing immediate, accurate financial information.

SECTION 2: PROJECT OBJECTIVES

The successful implementation of an Enterprise Resource Planning (ERP) platform requires clearly defined objectives that align technological solutions with institutional goals. The objectives outlined in this proposal are derived directly from the findings of the requirement-gathering exercise and reflect the operational, academic, administrative, financial, and strategic needs of TICH.

The proposed ERP platform is designed not merely to automate existing manual processes, but to transform institutional operations through integrated digital technologies, standardized workflows, improved governance, enhanced security, and data-driven decision-making. The objectives therefore focus on improving efficiency, accountability, collaboration, regulatory compliance, and service delivery across every functional area of the institution.

2.1 General Objective

The primary objective of this project is to design, develop, implement, and deploy a secure, scalable, and fully integrated Higher Education Enterprise Resource Planning (ERP) Platform that centralizes and automates the academic, administrative, financial, human resource, research, procurement, quality assurance, and operational functions of TICH while supporting institutional growth, regulatory compliance, and evidence-based decision-making.

The ERP platform will provide a unified digital environment through which students, staff, administrators, management, and external stakeholders can securely access the services relevant to their roles, thereby improving institutional efficiency, operational transparency, and overall service delivery.

2.3 Strategic Objectives

To achieve the overall project vision, the implementation will pursue the following strategic objectives:

1. Digital Transformation

Transform TICH from a predominantly manual and paper-based operational environment into a fully integrated digital institution where business processes are automated, standardized, and centrally managed.

2. Institutional Integration

Integrate all major institutional departments into a single centralized information system that enables seamless sharing of information while maintaining strict security and role-based access controls.

3. Operational Efficiency

Reduce administrative workload by automating repetitive tasks, accelerating approval workflows, and eliminating unnecessary manual processes.

4. Data-Driven Decision Making

Provide executive management with real-time dashboards, analytics, and institutional performance indicators to support strategic planning, monitoring, and policy development.

5. Regulatory Compliance

Ensure that all institutional operations comply with the Kenya Data Protection Act, Ministry of Education requirements, TVETA guidelines, Nursing Council regulations, KRA statutory requirements, and other applicable legal and regulatory frameworks.

6. Institutional Growth

Develop a scalable technology platform capable of supporting future institutional expansion, including additional campuses, community colleges, academic programs, mobile services, and external system integrations.

SECTION 3: PROJECT SCOPE

3.1 Introduction

The scope of the TICH Enterprise Resource Planning (ERP) Platform defines the functional, technical, operational, and organizational boundaries of the project. It specifies the business processes, institutional departments, user groups, system modules, technologies, integrations, deployment environments, and implementation activities that will be included in the project, while also identifying functions that are intentionally excluded from the current implementation phase.

The scope has been developed directly from the verified findings obtained during the requirement-gathering exercise. Every module, workflow, approval process, and user interaction included in this proposal reflects actual institutional operations observed across the Administration, Academics, Finance, Human Resource, Procurement, Quality Assurance, Community Colleges, ICT, Student Affairs, and Executive Management departments.

The objective of defining a comprehensive project scope is to establish a common understanding among all project stakeholders regarding the expected deliverables, implementation boundaries, and responsibilities throughout the project lifecycle.

3.2 Functional Scope

The ERP platform shall automate and integrate the following institutional business functions.

Departments and Modules

1. Administration Department

Institutional Budgeting & Strategic Planning

- **Multi-Tiered Budget Configuration:** System supports the creation, coordination, and oversight of institutional planning activities categorized across annual, monthly, and weekly timelines.
- **Cross-Departmental Financial Aggregation:** Consolidates monthly and annual budget requisitions pulled directly from all departments, including Finance, HR, Procurement, Infrastructure, Academics, Communications, ICT, Research & Monitoring and Evaluation, Networking and Marketing, and the SACCO.
- **Strict Requisition Deadlines:** Features built-in system constraints to prevent late submissions, mandating that all monthly departmental plans and budget requests must be presented by the middle of the prior month.
- **Curriculum System Adaptability:** Provides native architectural support for planning budgets under Competency-Based Education (CBE) frameworks alongside traditional educational systems.
- **Automated Approval Pipeline:** Integrates a structured workflow for plan verification and final fiscal clearance:
 - Departments submit plans to the Administration interface.
 - Plans are automatically routed to the Finance Department to verify the availability of sufficient funds.
 - Validated plans are escalated to the Executive dashboard (CEO) for final dual-authorization and signature by the TICH CEO and the Chairperson of the TICH-TRUST.
- **Digitized Fund Distribution:** Upon executive approval, the system triggers a fund release command through Finance, allowing Administration Department to digitally withdraw and distribute allocations to respective departments on a monthly basis.

Student Admissions & Lifecycle Onboarding

- **Dual-Access Application Portal:** Powers a fully online admission framework featuring multi-step digital forms (capturing personal information, academic history, supporting

documents, and program choices) accessible by external applicants or processed internally by the Admin office.

- **Onboarding & Bulk Enrollment:** Includes robust bulk enrollment utilities and self-service onboarding portals to process large student cohorts efficiently.
- **Digital Document Intake Checklist:** Enforces mandatory data and document uploads during student onboarding, securely archiving:
 - Full legal name and two colored passport photographs.
 - KCSE result slips or vocational qualifications (e.g., Class 8 certificates).
 - National ID copies or official Birth Certificates.
 - Digital receipt or proof of application fee payment.
- **Automated Admission Lifecycle Workflow:** Directs the progressive phases of student intake:
 - **Step 1:** Applicant submits the application, and the system instantly generates a Unique Application ID.
 - **Step 2:** The Academic Department reviews and verifies the qualifications within their dashboard.
 - **Step 3:** Upon academic approval, the system issues payment instructions where the Unique Application ID serves as the student's reference/account number for the application fee.
 - **Step 4:** Finance verifies payment manually, or the system automatically switches the status to "Paid" via real-time M-Pesa integration.
 - **Step 5:** Upon payment confirmation, the Administration Department generates the official admission letter, and the system produces a permanent Student Registration Number.
- **Structured Admission Letter Generation:** Automatically compiles and dispatches a comprehensive admission package to the student detailing:
 - Core course of study, chosen center of study, and official institutional reporting date.
 - **Structured Registration Numbers:** Formats codes based on campus parameters, such as THCM/N/J13-03/2026 (for Community Colleges utilizing course, exam body, month abbreviations, numbers, county codes, and year) or THCM/J13/2026.
 - **Automated Document Attachments:** Automatically appends relevant course-specific fee structures, institutional bank details, general enrollment requirements, and academically approved medical forms.

Compliance, Legal & Regulatory Oversight

- **Statutory Details Repository:** Centralizes and tracks the institutional compliance files, registration documents, and legal certifications required by oversight bodies.
- **Regulatory Alignment Alerts:** Monitors and evaluates operational alignment with the Ministry of Education, Kenya Revenue Authority (KRA), Public Health mandates, and technical examining bodies such as TVET and TVETA.
- **Inspection Readiness Dashboard:** Ensures all mandatory compliance documents are perpetually available online to facilitate seamless, on-demand physical or digital inspections by external legal and educational authorities.

Cashflow & Procurement Management Oversight

- **Procurement-to-Pay Lifecycle Tracking:** Provides administrative visibility over the institutional supply chain workflow:
 - Procurement identifies, vets, and approves an appropriate supplier.
 - The supplier delivers goods or renders services and submits an invoice to Finance.
 - Finance routes the invoice back to Procurement for absolute verification against delivery parameters.
- **Three-Way Match Verification:** Enables Procurement to confirm that physical goods or services match invoice details perfectly before returning the verified file to Finance; inaccurate invoices are automatically rejected back to the supplier.
- **Ledger Synchronization:** Once Finance executes payment via authorized digital methods (e.g., bank transfer or M-Pesa), the transaction is pushed directly into QuickBooks for synchronized financial reporting, audit trailing, and bank reconciliation.

Auxiliary Services & SACCO Facilities Operations

- **SACCO Housing Integration:** Integrates with the TICH SACCO data layer to manage institutional student hostels.
- **Hostel Awareness & Onboarding:** Automatically flags hostel availability to incoming students during the initial online admission process.
- **Automated Boarding Ledger:** Manages monthly recurring accommodation billings (configured at KES 3,500) and tracks individual student rent payments.
- **Cafeteria Operations Ledger:** Manages the operational parameters of the SACCO-run cafeteria, including tracking staff memberships and maintaining distinct accounting separation through a dedicated, isolated bank account.

2. Academics

The platform implements a role-based access management structure to isolate and enforce institutional oversight, policy updates, curriculum approvals, and resource scheduling across physical and virtual environments.

A. Academic Board Portal

The Academic Board Portal serves as the highest institutional policy engine on the platform, managing system configurations for quality assurance.

- **Policy Management:** Processes, passes and saves structural policy definitions, including pass marks, grading frameworks, and marking script compliance templates.
- **Verification Gates:** Provides final institutional verification for compiled course timetables, baseline fee structural changes, and moderated examination inventories.
- **Reporting Output:** Generates consolidated academic risk assessments and distribution metrics to deliver directly to the Overall Governing Board.

B. Academic Registrar Management Terminal

The Academic Registrar Management Terminal serves as the central control module for back-office student administration, catalog mapping, institutional scheduling, and student registration tracking. It executes administrative lifecycle logic and ensures data consistency across the university network.

1. Academic Calendar Configuration Engine

- **Dynamic Framework Curation:** The terminal houses a dynamic, flexible academic calendar builder designed to accommodate structural shifts in enrollment windows, holiday breaks, and practical field placement blocks.
- **The Primary Trimester Model:** The primary instance of the calendar engine enforces an academic year structure broken down into exactly **3 active terms/semesters per calendar year which can be changed if need arises. Specifies start sem dates and end dates.**
- **Intake Synchronization:** This scheduling engine directly synchronizes with the core institutional intakes framework to coordinate admission workflows for the cohorts e.g. January, May, and September cohorts, also specified by the Academic registrar

2. Entity Creation & Modification Workflow

The terminal provides structured administrative tools to initialize new structural units inside the global ERP ecosystem:

- **Departmental Initialization:** The Academic Registrar can provision new academic departments inside the master database.

- **Course Definitions Curation:** The Registrar creates and describes new courses, establishing permanent registry items containing prerequisite logic, core designations, electives, and specialization tracks.
- **CEO Authorization Gate:** To protect database integrity, newly created departments or course definitions are flagged as Inactive / Pending Sign-Off. The system restricts deployment until a final digital sign-off is completed via the Chief Executive Officer (CEO) account.

3. Student Lifecycle Administration Engine

This engine routes, processes, and archives student-initiated lifecycle modifications:

- **Waitlist Management Logs:** Automates student pipeline traffic by managing waiting lists, analyzing candidate distributions, and transitioning applicants to active status as enrollment spaces open.
- **Deferral Request Queues:** Evaluates and stores formal applications from students seeking to pause their studies, dynamically preserving their registration tokens across future terms while triggering financial pauses in the ledger.
- **Recognition of Prior Learning (RPL) & Credit Transfers:** Features a specialized ingestion interface to evaluate alternative entry paths and calculate transfer credits. It converts documented practical trades, work histories, or Community Health Provider (CHP) experience into accredited academic credit equivalents matching vocational frameworks.

4. Credential Verification & Exam

The Registrar coordinates the ultimate validation checks required to clear students for summative testing sequences:

- **Roster Compilation:** Processes data arrays to compile final, verified student rosters per active unit/module, serving as the baseline data for attendance logs.
- **Exam Eligibility Matrix Loops:** Runs a mandatory gatekeeping check across active student records before generating testing clearances:
 1. *Class Attendance Validation:* Verifies that the student's digitized check-in history satisfies a minimum threshold of exactly **90% attendance** across the unit duration.
 2. *Fee Clearance Evaluation:* Scans financial data tables to confirm an absolute **Zero Balance** status on tuition and institutional ledgers.
- **Official Examination Card Distribution:** Generates and distributes signed digital exam cards directly to compliant student portal profiles only after the record passes both the attendance and fee gates.

C. Heads of Departments (HODs) Dashboard

The HOD Dashboard provides course-building capabilities, curriculum curation interfaces, and instructor assignment management tools to section leads. It links high-level academic policies with day-to-day classroom execution.

1. Curriculum Design & Unit Customization Module

- **Draft Generation Workspace:** Grants department heads the tools to build, review, and adapt specific unit structures, operational modules, and technical courses.
- **Syllabus Parameterization Engine:** Requires HODs to explicitly define core unit criteria, including prerequisite rules, co-requisite tracks, elective configurations, and total learning contact hours per block.
- **Registry Verification Routing:** Completed module definitions are flagged as Pending Registry Verification and automatically escalated to the Academic Registrar Terminal for final system validation and course catalog activation.

2. Resource & Workload Balancer Matrix

- **Dual-Role Scheduling Configuration:** Because department heads often maintain teaching schedules while carrying out administrative tasks, the interface operates a unified resource balancer.
- **Allocation Oversight Engine:** Monitors contact hour distributions across faculty profiles. It cross-references active room schedules, lab usage parameters, and library asset bookings to prevent resource overallocation or calendar overlap.
- **Cap Safeguard Thresholds:** Automatically flags individual instructor profiles if cumulative assignments exceed weekly contact limits, alerting leadership to adjust assignments.

3. Departmental Segregation Matrix (The Three-Core Framework)

The platform isolates system controls, curriculum parameters, and diagnostic tools across three explicit departmental profiles:

Profile 1: Community Health and Development (CHD)

- **Regulatory Compliance Mapping:** Manages educational tracks and lesson configurations aligned with the Curriculum Development Assessment and Certification Council (CDACC) and Technical and Vocational Education and Training (TVET) frameworks.
- **Nursing Core Integration:** Operates as the primary administrative engine for the foundational **Nursing program**. It enforces statutory requirements from the Nursing Council across three core tracks: General Nursing, Midwifery, and Community Health Nursing.
- **Clinical Block Curation:** Bypasses standard semester timelines to manage a specialized block learning system (comprising six distinct milestones from the Introductory through the

Fifth Block). It locks student progression until clinical field logs and skills laboratory assessment marks match passing rules.

Profile 2: Vocational Training

- **Artisan Specialty Allocation:** Coordinates competency tracking matrices across five distinct artisan specialties:
 1. *Masonry*
 2. *Electrical Installation*
 3. *Welding*
 4. *Home Care Management (Community Colleges)*
 5. *Community Colleges Network*
- **NITA Framework Integration:** Uses the National Industrial Training Authority (NITA) modular framework rules. Instead of using traditional credit-hour metrics, the dashboard provides instructors with digital skill checklists to log practical competency development.
- **Decentralized Cell Monitoring:** Connects with the Affiliated Artisans Registry to track mentor assignments and practical field logs submitted from remote sub-county learning sites.

Profile 3: Information Communication Technology (ICT)

- **Internal Software Catalog Management:** Oversees internal technical certification paths, specifically governing the curriculum layers for the specialized **Data Science track** across Certificate and Diploma levels.

D. Teaching Staff (Trainers & Tutors) Workspace

This section defines the technical specifications and functional workflows for the shared instructional workspace. Designed as a unified interface, this workspace provides operational tools for active **Trainers and Tutors** while granting real-time monitoring, audit, and intervention capabilities to **Heads of Departments (HODs)** across both physical classrooms and virtual learning environments.

1. Lesson Plan Submission & Approval Pipeline

To maintain instructional quality across all departments, the platform enforces a digital state machine for lesson curation before any class session is initiated.

- **Structural Objective Compilation:** Tutors use a standardized digital template to input specific lesson objectives, planned topics, core competencies targeted, and total contact hours per module.

- **HOD Approval Dashboard:** Upon submission, the lesson plan is routed to the respective HOD's authorization terminal. HODs can review, edit, comment, or approve plans directly within the interface. Approved plans automatically clear the associated unit on the master academic timetable.
- **Historical Audit Repository:** All approved and modified lesson plans are indexed in a read-only database section accessible to the Academic Registrar to track institutional curriculum coverage.

2. Classroom Delivery & Environmental Interfaces

The ERP connects classroom tools with physical and digital assets to simplify delivery tracking for both instructors and departmental heads.

- **Virtual Learning System (VLS) Integration:** Provides single-sign-on (SSO) buttons to launch online lectures, track student login durations, and sync recorded video sessions back to the Student Portal.
- **Physical Asset & Space Registry Hooks:** Automatically cross-references physical timetables to reserve resources like the Central Library, specialized equipment hubs, and the ICT Computer Lab during scheduled class hours.
- **HOD Workload Allocation Monitor:** Feeds real-time delivery data into the departmental dashboard, tracking actual contact hours against assigned weekly targets to ensure balanced lecturer workloads.

3. Attendance Verification

To protect the integrity of the institution's exam eligibility gates, the platform uses a dual-layer verification process for student attendance logs.

Step-by-Step Attendance Workflow

1. **Sheet Generation:** The platform generates session-specific, printable attendance sheets bearing a unique system tracking ID for physical sessions.
2. **Physical Collection:** Students sign the physical paper sheet during regular class hours.
3. **Digital Roster Matching:** Tutors open the digital attendance register in their workspace, which is restricted to authorized personnel (tutor). They select checkboxes to match the platform's electronic roster with the physical signatures collected.
4. **Supporting Document Upload:** Instructors must use the mobile or web camera module to capture and upload a photo/image of the signed physical attendance sheet as an unalterable supporting document.
5. **HOD & Registrar Verification:** The uploaded image and matched digital roster are stored together in a secure ledger, allowing HODs and the Academic Registrar to do instant verification checks.

Attendance Risk Mitigation Matrix

The system automatically calculates individual student participation rates and applies visual status flags within the workspace, this is done per unit:

Calculated Attendance Bound	System Status Flag	Triggered Action / Portal Notification
90% and above Attendance	GREEN: COMPLIANT	Clears the student for exam card generation once fees are fully paid.
75% - 89% Attendance	AMBER: AT RISK	Pushes an automated warning alert to both the Student Portal, assigned tutor and the HOD panel.
74% and less at Term/Sem Close	RED: ELIGIBILITY BLOCK	Automatically locks the student's account out of the Exam Sitting Gatekeeper module.

4. Continuous Assessment & Performance Mapping Terminal

This terminal allows instructors and HODs to input and analyze student performance marks throughout the term.

- **Competency Spreadsheet Interface:** A direct data-entry grid for recording scores from continuous assessment tests (CATs), theoretical reviews, written assignments, and hands-on skills laboratory evaluations.
- **Automated Score Accumulation:** Individual marks are automatically compiled and weighted according to program-specific rules, building a cumulative score sheet for student profiles.
- **Departmental Performance Breakdown:** Generates analytical views for HODs, showing class score averages, failing trends, and practical completion rates across sub-county hubs. This gives leadership the insights needed to implement early learning interventions before final testing begins.

E. Curriculum Builders & Specialized Departmental Rules

The ERP uses a multi-tiered curriculum customization engine that accommodates external vocational training requirements, internal structured paths, and clinical block structures.

A. Core Vocational and Internal Formats

- **External Frameworks (NITA & CDACC/TVET):** Tracks modular trade progress using continuous competency checklists instead of traditional credit-hour equations. These configurations support localized artisan training tracks.
- **Internal Frameworks (Data Science):** Applies a semester-based credit matrix, tracking required contact hours, prerequisite sequences, core electives, and specialization tracks.

B. Specialized Nursing Department Configuration Rules

To maintain compliance with the Nursing Council guidelines, the platform enforces strict data gates for the Nursing program across the student lifecycle:

i. Ingestion Admission Filters

The portal applies automatic filtering rules during registration. An applicant account is blocked from choosing the Nursing track unless their verified secondary education certificate meets the following threshold:

Only accounts that pass this rule are categorized as eligible for enrollment in the program.

This will be mentioned in the website and will be verified by the Academic Registrar on the platform

ii. Academic Block System Framework

The Nursing curriculum operates within a specialized block learning system spanning **six sequential milestones**, with each block lasting **4 to 6 months**:

1. Introductory Block (*Requires immediate practical skills lab assessments before field routing*)
2. First Block
3. Second Block
4. Third Block
5. Fourth Block
6. Fifth Block

Students are locked out of regular semester-based timelines; instead, the system triggers a mandatory clinical training assignment at the close of each block.

This will be created by the Department head and approved by the academic registrar

iii. Assessment and Grading Weights Configuration

The evaluation engine compiles cumulative performance metrics by pulling from two separate sources, processing final marks according to specific criteria:

- **Written CATs Balance:** Accounts for **20%** of the total unit grading metric.
- **Skills Laboratory Evaluation:** Accounts for **10%** of the total unit grading metric.
- **Continuous Assessment Total:** Automatically locked at exactly **30%** of the final course grade.

The minimum passing bounds are managed via strict system validation logic:

- **Theory Examinations Passing Rule:** Programmed at a minimum of **60%**. Failing this grade automatically generates a Supplementary Theory Exam ticket.
- **Clinical Practical Evaluations Passing Rule:** Programmed at a minimum of **70%**. Failing this grade generates a Supplementary Clinical Assessment ticket.

3. Community College

This sub-module supports remote training setups, field registrations, and regional entrepreneurship tracking across decentralized networks.

A. Location & Campus Hierarchy Registry

The system structures institutional relationships using a hierarchical parent-child data array to organize regional campus entities:

Sub-Counties > Community Colleges > Students

- **Sub-County Hub Mapping:** Groups local branches under administrative regional boundaries (**Kisumu Central, Nyando, Muhoroni**) to align with local county entry parameters.
- **Facility Profiles Database:** Organizes attributes for remote centers, categorizing resource pools based on location types, such as **shared community halls, local church facilities, or mobile workshops**.
- **Affiliated Artisans Registry:** Registers local field experts, tracking their certifications to authorize them as practical mentors and trainers under NITA/CDACC regulations.

B. Offline-First Admissions & RPL Engine

- **Field-Agent Mobile Intake:** Provides low-bandwidth, offline-ready mobile web interfaces. Field mobilizers can input student details on-site during local drives; the application stores data locally and syncs with the central registrar database once a connection is established.
- **Recognition of Prior Learning (RPL) Engine:** Provides a non-standard entry evaluation workflow that maps alternative backgrounds, practical work histories, and Community Health Provider (CHP) status into vocational trade equivalence values.

C. Service-Linked Ecosystem & Work-Study Bridge

This ledger connects student academic milestones with local economic and community development projects.

- **Work-Study Stipend Ledger:** Tracks student field-service hours on regional engineering or sanitation assignments (e.g., bio-sanitation construction). It computes payment values or calculates tuition offsets to apply directly to student accounts in the finance ledger.

4. Scheduling, Timetabling & Lesson Plan Workflows

The platform provides synchronization tools to manage scheduling constraints across physical classrooms, virtual settings, and multi-campus locations.

A. Three-Matrix Timetable Generation Workspace

The system core automatically constructs three separate, data-validated matrices to prevent calendar conflicts across facilities:

- 1. **Class & Tutor Matrix:** Validates instructor availability against required unit contact hours and total learning hours per module.
- 2. **Annual & Term Master Schedule:** Maps out enrollment windows, intake points (January, May, September), holiday breaks, and practical field placement blocks.
- 3. **Exam Schedule & Invigilator Directory:** Assigns physical evaluation spaces, lists invigilators, and checks for resource over-allocations.

B. Lesson Plan Authorization State Machine

Tutors must route their lesson outlines through a standard approval workflow before conducting classes. The system tracks lesson progress through four distinct states:

5. End-to-End Examination Lifecycle & Assessment Engine

This framework governs the academic evaluation pipeline, managing paper submissions, verification routines, eligibility checks, and post-result adjustments.

A. Master Cumulative Weight Calculations

Final course units are calculated using an automated assessment processor. The system combines multiple input categories based on specific weight rules:

Final Unit Score = Attendance Log (5%) + CATs / Practical Tasks (Variance) + Final Summative Exam (Variance)

Evaluation Component	System Tracking Parameter	Mandatory Weight
Classroom Attendance	Verified session check-ins via localized sign sheets.	5% (Fixed Minimum Constant)
Continuous Assessment (CATs)	Theoretical tests, review quizzes, and assignments.	Program Dependent Variance
Practical Evaluations	Skills lab work, trade mastery tests, and field exercises.	Program Dependent Variance
Final Term Examination	Summative theoretical and practical assessments.	Program Dependent Variance

B. Examination Validation & Moderation Lifecycle

For internal curriculums, evaluation papers move through a multi-step moderation workflow:

1. **Draft Upload:** The module tutor writes the draft exam questions and uploads the file to the portal.
2. **Tabled Collection:** The Academic Registrar organizes submissions and tables them for review.
3. **Board Moderation:** The Academic Board holds a review meeting to modify and approve assessments to verify quality standards.
4. **Secure Encryption:** Approved papers are compiled and encrypted by the Registrar for secure delivery to examination rooms.

C. Gatekeeper Eligibility Verification Rules

Before generating an individual student's exam card or granting room entry, the platform executes a check across four operational parameters:

- **Class Attendance Rule:** Confirms that the student's cumulative participation log 90\% across the unit duration.
- **Fee Clearance Rule:** Evaluates the billing ledger to confirm a Zero Balance status on tuition accounts.
- **Invigilator Status Flag:** Verifies that an approved invigilator is assigned to the exam session.
- **Exam Card Generation:** Confirms that a valid, system-signed exam card has been issued to the student portal profile with a unique examination number and QR code.

Note: The platform provides a Student Complaints Portal to allow learners to appeal eligibility flags or report processing errors before testing blocks close. Following exam completion, invigilators upload scripts directly into the Examination Officer's ledger.

D. Post-Result Academic Modification Pathways

Once the Academic Board verifies and publishes final grades, the system manages alternative performance paths via three distinct workflows:

Special Examinations Workflow: Available for students who missed original testing sessions due to documented extenuating circumstances. Requests are routed to the Academic Registrar for review and scheduling.

1. **Supplementary Examinations Workflow:** Triggered when a student fails to hit the minimum passing mark for their specific program track. This path requires joint verification from the Academic Registrar and the Finance Department to log and confirm the processing fee payment.

2. **Remarking Application Pipeline:** Initiated when a student formally contests their published result. The system flags the original script, assigns it to an independent external examiner, and updates the grade ledger following Academic Board validation.

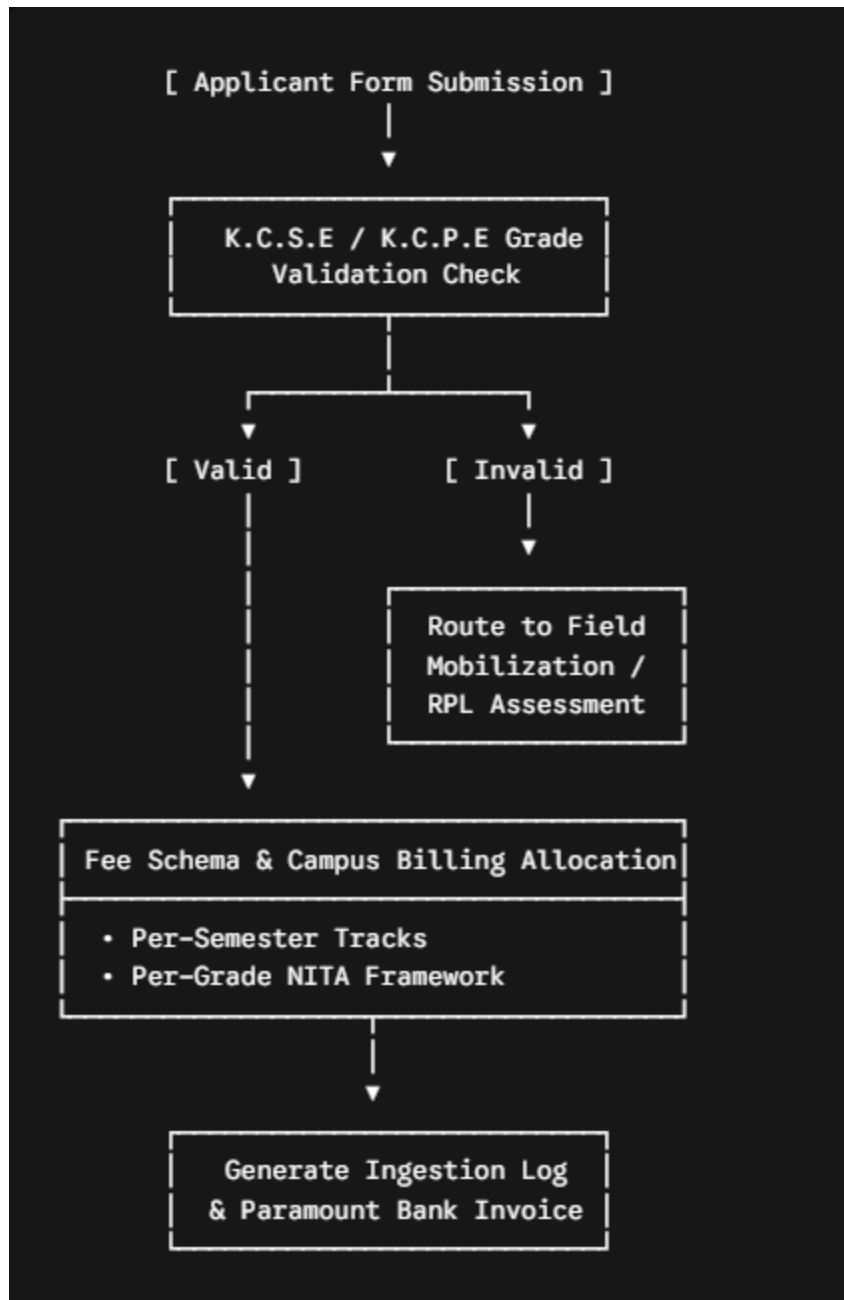
6. Cross-Module ERP Integration Matrix

To maintain data consistency, the Academics & Admissions Module shares real-time data lookups with other components of the platform through dedicated communication nodes:

Targeted ERP Module	Shared Data Elements & System Links	Technical Purpose & Validation Use
Finance Department	Fee structures, payment flags, paid supplementary fees, and work-study offsets.	Enforces the Zero-Balance Check before generating exam cards and processes tuition balance deductions from work-study logs. The balance percentage is editable within the platform
Human Resources (HR)	Faculty contact hours, lesson plan completion metrics, and instructor workload logs.	Balances weekly teaching assignments and matches instructor fields with required unit profiles.
Marketing & Mobilizers Portal	Public interest profiles, target sub-county center selections, and referral records.	Tracks regional community outreach performance and converts prospective leads into student accounts at the admissions desk.
TICH SACCO & m-Fundi Network	Completed certification lists, student ID fields, and loan requests.	Syncs graduate cohorts with the m-Fundi service directory and runs credit checks for toolkit or startup financing.

Admissions Validation Rules & Program Matching Engine

To protect downstream grading workflows, the platform utilizes strict ingestion validation logic when processing submissions from both the online Portal and field-agent devices.



3. Human Resource

To support the administrative backbone of both the central Kisumu campus and the decentralized network of sub-county community colleges, the **Human Resource & Payroll Module** acts as a unified engine. It automates the complete employee lifecycle, ensures strict compliance with Kenyan statutory regulations, and seamlessly interfaces with the Employee Self-Service (ESS) platform.

1. Core Architectural Modules

A. Full Staff Lifecycle Management

- **Recruitment & Onboarding Pipeline:** Allows HR to transition selected applicants directly into the staff database. The onboarding interface captures employment category (Permanent, Contract, Intern), department assignment, position, and monthly gross pay.
- **Dynamic Template Engine:** Houses standardized system templates for employment contracts, probation evaluations, transfer letters (e.g., moving staff between sub-counties), and exit clearing checklists.
- **Contract Lifecycle Monitor:** Tracks contract end-dates and automatically triggers system notifications and emails to HR and HODs exactly **1 month prior** to contract expiry to initiate renewal or transition protocols.

B. Kenyan Compliant Payroll & Compensation Engine

- **Statutory Deduction Processor:** Automatically calculates and deducts PAYE, NSSF (Tier I & II), SHA (Social Health Authority / legacy NHIF), Affordable Housing Levy (AHL), and voluntary pension schemes from gross pay using updated Kenya Revenue Authority (KRA) and government tax bands.
- **Automated Disbursement & Banking Matrix:** Captures comprehensive banking details (Account Name, Account Number, Bank, Branch Code) and compiles monthly electronic bank transfer files (EFTs/PESTs) overseen and authorized by the Finance Department.
- **Tax Auditing & P9 Center:** Compiles monthly statutory reductions into an annual cumulative **P9 Form**, accessible to employees via the ESS portal for direct KRA iTax returns filing.

C. Payroll

Onboarding Process

- Log in to HR/Finance dashboard
- Onboard an Employee: Enter name (or select from site db), details, select category (Contract, Permanent, Intern)
- Include gross pay for the employee (Monthly)
- Select the department in which they fall into, if they fall into any
- Include their position in that department.
- Enter bank account details (Account number, Account name, Bank branch etc.)
- Enter Statutory details

Calculation of deductions

- Payments are remitted to employees minus deductions as stipulated by KRA and the Government of Kenya

- Payments are automated and overseen by the Finance department.
- The dashboard should also have P9 (audit of deductions for an employee annually)

D. Time, Attendance & Leave Management

- **Leave Track Workflow:** Digital application portal for multiple leave categories (Annual, Sick, Maternity, Paternity, Compassionate). It automatically computes running balances (leave days) and routes requests to HODs and HR for authorization.
- **Attendance Registry:** Tracks clock-ins and field logs, feeding directly into the payroll engine to audit contract hours, shift compliance, and work-study field assignments.

E. Compliance, Performance & Disciplinary Dashboard

- **Professional Development Tracker:** Logs staff academic qualifications, certifications, skills training history, and continuing professional development (CPD) credits.
- **License Expiry Registry:** Monitors valid professional operational licenses (e.g., nursing or clinical officer licenses for health tutors) with early warning alerts prior to expiry.
- **Disciplinary Record & Policy Hub:** A secure dashboard containing incident registries, warning letters, and disciplinary hearing notes. It includes a mandatory **Policy Acknowledgment Interface** where staff must digitally sign off on revised institutional policies or compliance directives.

2. Role-Based Access Control (RBAC) Matrix

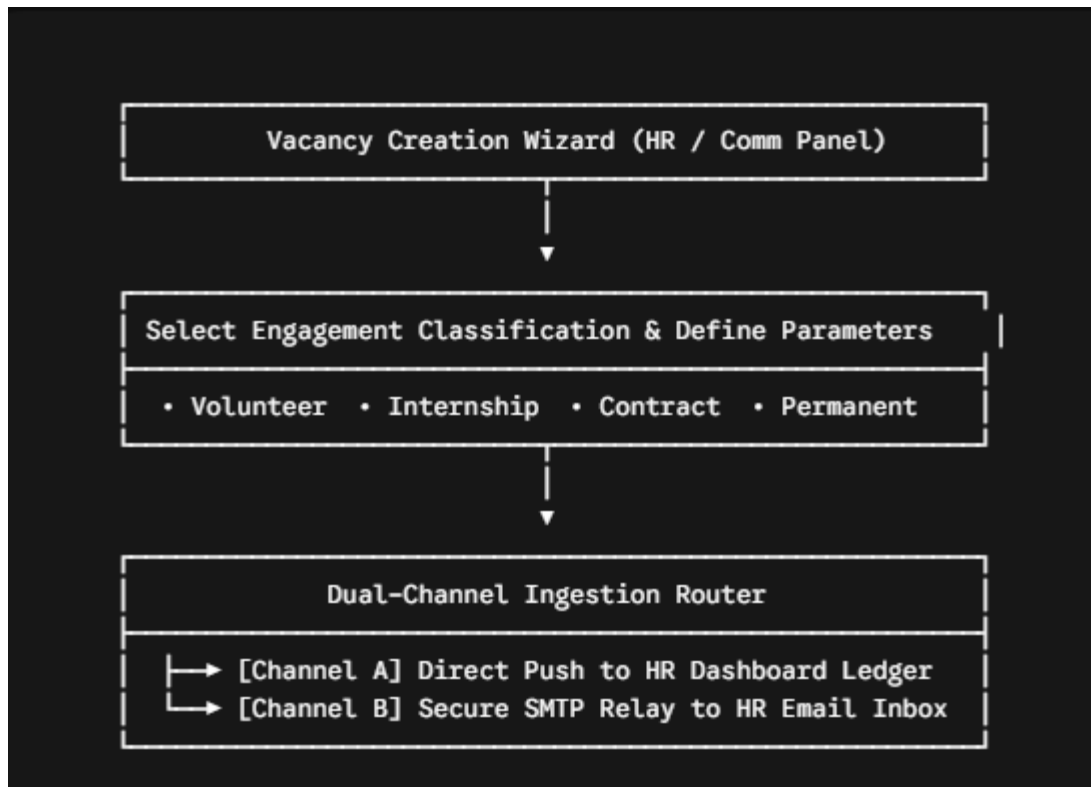
To preserve sensitive data privacy (such as gross salaries, banking information, and disciplinary actions), system interactions are restricted based on organizational standing:

Module Feature	Regular Staff / Tutor	Head of Department (HOD)	HR Manager	Finance Officer
My Profile Bio-Data / P9 Access	Read / Request Update	Read Only (Team)	Full Access	Read Only
New Employee Onboarding	No Access	No Access	Full Access	View & Verify Pay

Salary & Deductions Settings	No Access	No Access	Read Only	Full Access (Edit)
Leave Management	Request & View Balance	Approve / Reject Team	Full Access & Overwrite	View Only
Disciplinary & Policy Upload	View & Digitally Sign	View Team Incidents	Full Access (Create)	No Access
Contract Expiry Alerts	View Notification	View Notification	Full Access (Ma	

Communications Job Posting Panel

This portal grants administrative access to both the **HR Department** and the **Communications Department** to create, schedule, edit, and archive vacancies displayed on the public website.



A. Vacancy Creation Wizard Parameters

When staging a new opening, the poster must configure the engagement classification rules through a structured dropdown matrix:

Engagement Classification	Mandatory Metadata Requirements	System-Enforced Constraints
Volunteer	Project/Department Tag, Scope of Work	No duration cap; defaults to open-ended.
Internship	Primary Mentor Assignment, Duration Field	Duration must be specified in months (e.g., 3 Months, 6 Months). Triggers the dynamic internship promotional text on the frontend.

Contract	Salary/Stipend Scale, Duration Field	Requires hardcoded Start and End Dates; triggers automated contract renewal alerts 30 days prior to expiry.
Permanent	Salary Scale, Grading Tier, Benefits Allocations	Subject to final executive board approval flags before deployment.

B. Publishing Permissions

- **HR Department:** Holds full administrative rights over all job categories, including the modification of internal salary/stipend scales, grading tiers, and internal contract parameters.
- **Communications Department:** Holds publishing rights to structure, format, publicize, and clean up public-facing text, ensuring alignment with institutional brand standards before going live.

Application Routing & HR Dashboard

To prevent data loss and ensure rapid processing, the application submission engine executes a **dual-channel ingestion protocol** upon every successful public submission:

Channel A: [Secure SMTP Relay \(Email Inbox Integration\)](#)

The system packages the applicant's text inputs into a structured HTML template, attaches all uploaded documentation (CV, cover letters) as verified binary files, and routes the package directly to the **HR Department's primary email inbox**.

Channel B: [HR Application Ledger Dashboard](#)

Concurrently, the data is saved to the ERP's internal database, generating a new interactive row within the backend **HR Applications Dashboard**. This ledger gives HR officers tools to manage the applicant pipeline:

[New Submission] → [Reviewing] → [Interview Scheduled] → [Accepted / De

- **Inline Preview Engine:** Allows HR personnel to open, read, and evaluate an applicant's uploaded CV and credentials directly inside the dashboard workspace without downloading files locally.

- **Verification Actions:** Features quick-action triggers to email applicants directly from the platform, log interview schedules, append evaluation notes, and seamlessly transition selected candidates into the active employee or work-study roster.

Cross-Departmental Enterprise Integration Matrix

Linked Department	Specific Intersecting Areas & Data Shared	System Function & Operational Rule Validation
Communications Department	Public Vacancy Text Elements Marketing Channel Telemetry Active Job Status Flags	Grants the communications desk editing and styling privileges over active public vacancies to maintain consistent institutional messaging.
Human Resources (HR)	Central Applications Ledger Inbound Attachment Repositories Vacancy Parameter Matrix	Serves as the primary operational terminal for scheduling interviews, evaluating CV uploads, monitoring contract timelines, and archiving submissions.

4. Finance & Fees

Student Accounts & Accounts Receivable (AR)

- **Flexible Fee Configuration:** Financial administrators can configure comprehensive fee structures categorized by academic program, academic year, specific module/unit, or distinct student classifications.
- **Automated Invoice Generation Engine:** System automatically computes and generates itemized billing invoices for tuition fees, application fees, supplementary examinations, and graduation fees.
- **Billing Workflow Automation: Student Billing Records:** The billing system pulls student data using their unique Registration Number, full name, class, and course.
 - **Standardized Invoice Ledger:** Invoices are automatically stamped with the generation date and a structured chronological number using the format [Student Registration Number] - 001.
 - **Omni-Channel Dispatch:** Finalized invoices are pushed directly to the Student Portal and delivered to the student's registered email address.
- **Real-Time Payment Settlement:** Integrates an instant clearing framework where student-initiated payments reflect on the Student Portal and the Financial Administration Dashboard immediately, triggering an automated email payment confirmation.
- **Multi-Gateway Payment Integration:** Seamlessly supports payment processing through M-Pesa as the primary channel, with robust architectures supporting direct bank integrations, mobile money alternatives, and credit/debit card processing.

General Ledger & Financial Reporting

- **Centralized Treasury Operations:** Features a single Main Account architecture to manage and reconcile all institutional financial transactions.
- **Automated Accounting Integrations:** Processes financial transactions and ensures systemic entry into a QuickBooks like dashboard within the platform to power underlying accounting and financial reporting streams.
- **Live Dashboards & Compliance Audits:** Real-time financial dashboards providing end-to-end audit trails for accounts receivable, accounts payable, and asset management.
- **Statutory Tax Engine:** Complete alignment with the Kenya Revenue Authority (KRA) guidelines for seamless institutional tax compliance and automated generation of localized tax returns.
- **Standard Financial Statement Suite:** Automated, on-demand compilation of compliance-ready financial reports, including:
 - Balance Sheets.

- Trial Balances.
- Profit and Loss Accounts (Income Statements).
- General Ledgers.
- Cashflow Statements.

Project, Grant & Donor Fund Accounting

- **Donor-Funded Project Management:** Tracking systems to monitor independent institutional activities and separate project expenses.
- **Grant Budgeting & Forecasting:** Intake tools to absorb cost estimations and itemized project budgets submitted by project leaders.
- **Donor Invoicing & Drawdowns:** Specialized billing modules to generate and route institutional invoices directly to external donors and sponsors.
- **Multi-Currency Conversion Engine:** Built-in forex handling to receive donor disbursements in foreign denominations (specifically USD) and automatically track conversion into local currency (KES) within the central main account.
- **Grant Accountability Reporting:** Specialized reporting tool to construct and export post-activity financial utilization reports for donors to guarantee transparency and fiscal accountability.

Institutional Payroll & Statutory Compliance

- **Integrated HR-Finance Portal:** Shared dashboard access enabling financial officers to seamlessly manage the employee payroll lifecycle.
- **Employee Payroll Onboarding:** Allows administrators to register staff profiles directly or select existing personnel records from the institutional database, classifying them into specific employment tiers like Contract, Permanent, or Intern.
- **Departmental Cost Centering:** Assigns personnel records to their respective departments and structural organizational positions to maintain strict budget separation.
- **Compensation & Banking Registry:** Captures definitive gross monthly salary allocations alongside comprehensive banking data, including account numbers, account names, and bank branch routing codes.
- **Automated Statutory Deductions Engine:** Calculates and applies localized statutory deductions; including PAYE, NHIF, NSSF, and applicable housing levies or pension allocations; conforming precisely to current Kenyan government mandates.
- **Disbursement Oversight:** Automated net salary remittance engines supervised and authorized by the Finance Department.
- **Tax Document Self-Service:** Automated annual compilation of statutory P9 certificates for each employee to ease personal tax filing audits.

Accounts Payable (AP) & Supplier Management

- **Vendor Account Provisioning:** Creates dedicated billing and account tracking profiles for verified suppliers based on procurement parameters.
- **Three-Way Matching & Invoice Verification:** Establishes a security checkpoint where incoming supplier invoices are routed across systems:
 - Procurement initiates and approves budgets and suitable suppliers.
 - Suppliers submit invoices to the Finance Department.
 - Finance routes the document back to Procurement to confirm the successful physical delivery of goods or execution of services.
- **Approval & Rejection Workflows:** Verified invoices move forward to final payment processing, while disputed or inaccurate invoices are automatically rejected and routed back to the supplier for corrections.
- **Disbursement & Reconciliation Execution:** Processes payments via authorized methods such as direct bank transfers or corporate M-Pesa channels, storing systemic documentation for seamless subsequent bank reconciliations and external audits.

5. Students' portal

Identity, Authentication & Profile Management

- **Secure Access Gateway:** Students securely authenticate and log into the portal using their unique Registration Number and password. Password reset will be available incase a student forgets their password. Will be limited to 3 times in 7 days. If it surpasses that, the issue is escalated to the ICT/ Technical team
- **Multi-Factor Security:** Access is reinforced via an optional multi-layered security architecture, including Two-Factor Authentication (2FA) delivered via SMS, WhatsApp, Email, or an Authenticator App.
- **Profile Self-Service:** Students can view their comprehensive personal bio-data profile and dynamically update permitted fields directly through the dashboard.
- **Session Lifecycle Control:** Secure logout functionality allows students to terminate active sessions reliably to maintain data privacy.

My Profile - "the digital me"

The student can view:

- Full name, student ID, programme of study, year of study, current semester, registration number
- Date of birth, gender, nationality, ID/passport number

- Phone number, email, postal address, home county
- Next of kin / emergency contact
- Profile photo (upload)

Some fields are editable by the student directly; others (name, ID number, date of birth) require the registry office or eligible office to approve the change.

Academic & Learning Management System (LMS)

- **Academic Calendar & Programs:** Access to detailed programmatic timelines, active academic calendars, and specific semester or block training structures.
- **Self-Service Unit Registration:** Online enrollment capabilities enabling students to register for their required courses and units within the active institutional registration window. This can also be done by the HOD & Academic Registrar when creating semester and annual schedules.
- **Personalized Timetables:** Real-time visibility into registered units alongside personalized weekly class and lecture schedules.
- **Digital Courseware Repository:** Centralized download center for accessing learning materials uploaded by lecturers, including notes, presentations, instructional videos, and reference publications.
- **Virtual Classroom Portal:** Direct access to attend live virtual classes and retrieve recorded lecture archives on demand.
- **Attendance Monitoring Tracker:** Live dashboard displaying detailed classroom attendance logs and cumulative attendance percentages.
- **Collaborative Communication Hub:** Interactive features to participate in online academic discussions, receive global institutional announcements, and message lecturers directly.
- Nursing students are required to pay 5% of total amount of school fees for semester course registration, other course there is no fee check for course registration in a start of a semester

Assessments & Examination Management

- **Assessment Schedules:** Centralized calendar showing Continuous Assessment Test (CAT) schedules and final examination timetables.
- **Online Academic Appraisals:** Digital assessment interface for completing online quizzes and scheduled coursework examinations.
- **Performance Grade Book:** On-demand access to verified CAT scores, skills laboratory/practical assessment results, and final examination marks once officially published.

- **Progress Analytics:** Visual tracking metrics to monitor long-term academic progress, credit completions, and cumulative grades.
- **Exam Eligibility Checker:** Automated compliance check that cross-references institutional requirements (e.g., verifying the mandatory 90% class attendance threshold and zero-balance fee clearance) to establish exam eligibility.
- **Digital Examination Card Generation:** Automated download portal for securing official examination cards once all academic and financial requirements are cleared.
- **Grades / Results:** At the end of each assessment (test, assignment, exam), the lecturer enters marks into the system. The student sees them appear in the portal:
 - **Coursework marks** - quizzes, assignments, CATs (continuous assessment tests)
 - **Exam mark** - final exam, entered after the exam period
 - **Final grade** - the official letter grade or percentage after the lecturer approves the
- **Transcript:** A cumulative, official-grade record of every course the student has taken, the credit hours, the grade, and the credit earned. The student can view results on the platform at any time. The official transcript is generated by the registrar (often with a watermark, and signature).
 - **The workflow (Transcript):**
 - 1. Log in, click **Academics > Transcript**.
 - 2. See a list of all units/modules taken, with marks per assessment and final grade.
 - 2. Student view the results but cannot download.
 - 3. Request an official version > routes to the registrar's queue.
 - 4. The registrar generates, signs, and either emails a sealed PDF or hands the student a printed copy.

Financial Services & Fee Management

- **Real-Time Financial Statements:** Complete visibility into personal fee balances, itemized invoices, payment histories, outstanding arrears, and downloadable payment receipts.
- **Integrated Payments Engine:** Self-prompted fee payment capabilities supporting automated invoicing for tuition, application fees, supplementary exams, and graduation fees.

- **Automated Clearing & Confirmation:** Direct integration with primary payment gateways (M-Pesa and bank networks) providing instantaneous payment reflection on the portal and automated email confirmations.

Walkthrough

1. Log in, click **Finance > My Statement**
2. See the current semester's invoice - itemized.
3. See the running total paid + balance.
4. See each payment with its **receipt number, date, and method** (M-Pesa).
5. Download or print the statement (useful for parents / sponsors).
6. When the balance hits zero and the course is completed, the financial clearance flag flips on.

Campus Services & Auxiliary Support

- **Lifecycle Requests Workflow:** Electronic submission portal for official institutional requests such as academic deferments, course withdrawals, or program readmissions.
- **Quality & Lecturer Evaluations:** Secure and structured interface to complete mandatory lecturer and course evaluation forms during designated institutional assessment windows.
- **24/7 AI Chatbot Assistant:** Multi-lingual virtual assistant (English and Kiswahili) accessible via the web and mobile interfaces to handle instant inquiries regarding admissions, fee balances, and timetables.
- **Dynamic Notification Center:** Omni-channel notification system delivering automated alerts for upcoming assignment deadlines, fee reminders, examination schedules, sudden timetable modifications, and general academic updates.

Graduation Clearance & Registry Operations

- **Official Document Center:** Formal request portal for provisioning certified transcripts, institutional recommendation letters, completion letters, and graduation clearance forms.
- **Multi-Department Clearance Tracker:** Real-time progress tracker monitoring mandatory final-year milestones across independent nodes, specifically Finance, Library, Hostels, Academic Departments, and the Registrar.
- **Tamper-Proof Credential Downloader:** Secure download vault for lifetime access to approved digital transcripts, graduation certificates, and authenticated academic documentation embedded with secure metadata.

6. Staff Portal

Academic & Learning Management (For Tutors/Lecturers)

This module allows teaching staff to manage their day-to-day classroom and administrative responsibilities seamlessly.

- **Lesson Plan Manager:** A dedicated space for tutors to create and submit lesson plans prior to classes for HOD or Academic Registrar approval.
- **Digital Attendance Register:** Tools for tutors to log, sign, and track student attendance per class, with automated alerts when a student risks dropping below the 90% exam eligibility threshold.
- **Continuous Assessment & Grading Hub:** Interface to input scores for written CATs, assignments, practicals, and skills laboratory assessments. It should automatically compile these into cumulative score sheets.
- **Learning Content Repository:** A dashboard for uploading course notes, presentations, videos, and reference documents directly to the Student Portal.
- **Virtual Classroom Integration:** Links to launch physical or online classes and manage recorded lectures.

Faculty & Leadership Dashboard (For HODs & Academic Registrar)

Designed for academic oversight, tracking program development, and structural approvals.

- **Curriculum & Modular Builder:** Tools for HODs to create units/modules per course and submit them to the Academic Registrar for final approval.
- **Workload Balancing & Allocation:** A matrix dashboard to monitor lecturer workloads and assign units to specific tutors at the start of the semester.
- **Collaborative Timetable Builder:** A shared workspace for the Academic Registrar and HODs to construct annual, semester, and exam timetables.
- **Exam Lifecycle & Moderation Hub:**
 - Allows tutors to submit draft exams to the Registrar.
 - Provides a workflow for the Academic Board to review, moderate, and approve exams.
 - Enables the Registrar to approve special or supplementary exam requests forwarded by students.
- **Admissions & Credit Evaluation Workspace:** For reviewing student applications, handling waitlists, deferrals, and evaluating transfer credits.

Employee Self-Service (ESS) (For All Staff Members)

A centralized hub for personal administrative management, ensuring staff don't have to visit HR manually for routine tasks.

- **My Profile & Bio-Data:** View and request updates for personal details, banking information, and statutory numbers (PAYE, NHIF, NSSF).
- **Payroll & Tax Center:** Secure access to download monthly digital pay slips and annual P9 forms for KRA tax returns.
- **Leave & Time-Away Management:** Interface to apply for various leaves, check remaining leave balances, and track HR approvals.
- **Performance & Appraisals:** View assigned Key Performance Indicators (KPIs), submit self-appraisals, and track professional training history.
- **Policy & Compliance Acknowledgment:** Dashboard to view and digitally sign off on institutional policies, disciplinary updates, or license expiries.

7. Quality Assurance

Quality Assurance & Compliance Module

The Quality Assurance (QA) & Compliance Module serves as the institutional auditing framework within the TICH ERP. This module provides the **Quality Assurance Officer** with centralized administrative tools to deploy quality plans, enforce regulatory compliance, and evaluate structural indicators. Concurrently, it gives **Heads of Departments (HODs)** decentralized interfaces to complete internal audits directly within the system.

QA Officer Command Center

The QA Officer Dashboard controls quality tracking, staff training logs, and regulatory compliance registers across all physical facilities and decentralized sub-county hubs.

- **Quality Control Oversight:** Coordinates the generation and system-wide roll-out of targeted quality assurance plans. The interface monitors the execution of training programs, lesson plan submission cycles, and the delivery of learner support networks.
- **Capacity Building Registry:** Manages internal capacity-building logs and schedules staff training sessions. These tools support faculty in understanding quality standards, checking attendance parameters, and identifying operational bottlenecks before final exam windows.
- **Unalterable Audit Logging:** Automatically logs identified quality exceptions, tracks resolution milestones, and compiles historical data into periodic Quality Level Reports for review by executive management and the CEO.

Interactive Assessment Sheet Orchestration Engine

This dedicated sub-panel allows the QA Officer to build, distribute, track, and compile digital evaluation forms across the institution's departments without relying on external files or paper processes.

A. Dynamic Form Builder & Dispatch Matrix

The QA Officer can use an internal form builder to create custom evaluation metrics tailored to each department's unique rules:

- **CHD & Nursing Track Parameters:** Focuses on Nursing Council entry rule validation
- **Vocational Training Parameters:** Evaluates modular NITA trade progress, practical competency checklist records, and local artisan mentor logs.
- **ICT Department Parameters:** Tracks software certification updates, computer lab hardware downtime, and library resource availability metrics.

B. Assessment Lifecycle State Machine

The platform manages the progress of dispatched evaluation sheets through a secure sequence of states:

C. Departmental Execution & Submission Interface

When an assessment sheet is dispatched, it appears as an outstanding task on the respective HOD's and department members dashboard.

- **Direct Data Ingestion:** HODs fill out the evaluation criteria directly within their workspace, drawing from current student performance sheets, tutor workload metrics, and lesson approval logs.
- **Evidence Linking Validation:** The interface requires the HOD to link supporting digital evidence; such as sample attendance sheets with photo verifications or moderated exam drafts, file uploads and attachments etc. before the system allows final submission.

Distributed Departmental Self-Audit Framework

To foster accountability, the system requires each department head to conduct periodic internal quality audits using parameters set by the QA Officer.

Targeted Department Panel	Evaluated Component	Mandatory Audit Validation Rule

Community Health & Nursing	Clinical Practice & Grading	Confirms that the minimum pass mark for clinical practical examinations is locked at $\geq 70\%$, and checks that theory grades match the 60% threshold.
Vocational Training	Competency Tracking	Verifies that NITA-registered artisan students are assessed using practical checklists rather than traditional written tests.
Information Technology	Resource Allocation	Cross-checks physical Computer Lab usage logs against student rosters to prevent overcrowding and ensure sufficient workstation access.
Central Admissions Desk	Ingestion & Verification	Audits registration pipelines to ensure all active Nursing students meet minimum K.C.S.E cluster grades (C in English/Kiswahili, C in Biology, C+ in science).

Analytics, Reporting, & Executive Feedback Loops

Once assessment sheets are returned from the departments, the central compilation engine automatically calculates performance metrics.

Management Feedback Loop

- **Deficiency Identification:** If a department's compliance score falls below preset limits, the platform generates a Quality Corrective Action flag. This flag locks downstream module creation tools until the department logs a resolution plan.
- **CEO Report Delivery:** The platform compiles these scores into clean, visual summaries. These are sent directly to the Academic Board and the Chief Executive Officer (CEO) to support long-term planning and confirm compliance with legal and educational standards.

8. Procurement

The Procurement Department is responsible for managing the acquisition of goods, services, and institutional assets required to support the operations of TICH. The department ensures that all purchases are conducted transparently, competitively, and in accordance with institutional procurement policies and financial controls.

The requirement-gathering exercise established that procurement activities involve requisition management, supplier sourcing, quotation evaluation, supplier selection, invoice verification, payment processing, goods receipt, inventory management, and asset tracking. The proposed ERP system shall automate and integrate these processes to improve accountability, efficiency, compliance, and reporting.

Requisition

1. Requisition Initiation

The procurement process begins when a department identifies a need for goods, services, equipment, or supplies.

The requesting department shall:

- Create a procurement requisition.
- Specify the required items or services.
- Indicate quantities required.
- Provide justification for the request.
- Submit the requisition for approval.

The system shall generate a unique requisition number for tracking purposes.

2. Requisition Review and Approval

Upon submission, the requisition shall undergo the institution's approval process.

The system shall support:

- Departmental approval workflows.
- Budget availability verification.
- Management approvals where required.
- Requisition status tracking.

Only approved requisitions shall proceed to supplier sourcing.

3. Supplier Quotation Collection

After approval of the requisition, the Procurement Department shall invite quotations from multiple suppliers.

The procurement team shall:

- Request quotations from qualified suppliers.
- Receive quotation documents.
- Record supplier details.
- Upload supplier quotations into the system.
- Maintain a history of all submitted quotations.

The system shall allow attachment and storage of quotation documents for future reference and auditing purposes.

4. Procurement Committee Evaluation

The requirement-gathering exercise established that supplier selection is conducted through a Procurement Committee.

The committee shall:

- Review all submitted quotations.
- Compare supplier pricing.
- Evaluate product quality.
- Assess supplier reliability.
- Consider delivery timelines.
- Verify supplier compliance requirements.

The system shall support:

- Quotation comparison.
- Supplier evaluation scoring.
- Committee recommendations.
- Approval documentation.

5. Supplier Selection

After evaluation, the Procurement Committee shall identify the most suitable supplier.

The system shall:

- Record the selected supplier.
- Document reasons for selection.
- Maintain records of unsuccessful quotations.

6. Invoice Verification

Once a supplier has been selected, the supplier shall issue an invoice.

The Procurement Department shall verify that:

- The invoice corresponds to the approved requisition.
- Quantities match the requisition.
- Prices match the approved quotation.
- Approved procurement procedures were followed.

The ERP system shall perform a three-way verification process involving:

1. Approved Requisition.
2. Approved Supplier Quotation.
3. Supplier Invoice.

7. Payment Processing

After successful invoice verification, the procurement department shall forward the approved invoice to the Finance Department for payment processing.

The system shall:

- Route verified invoices to Finance.
- Track payment approval status.
- Record payment references.
- Update procurement records once payment is completed.

Procurement personnel shall be able to monitor the payment status without requiring direct intervention from Finance.

8. Delivery of Goods or Services

After payment approval, the supplier shall deliver the goods or services.

The delivery may occur through:

- Supplier delivery to the institution.
- Collection of goods by authorized procurement personnel.

Upon receipt of goods, the Procurement Department shall verify:

- Correct quantities delivered.
- Correct specifications supplied.
- Acceptable quality standards.
- Compliance with the approved order.

The system shall record:

- Delivery confirmation.
- Date of receipt.
- Receiving officer.
- Delivery documentation.

- Any shortages or damages identified.

The supplier shall provide a delivery note upon delivery.

The system shall:

- Store delivery note records.
- Link delivery notes to the original requisition.
- Link delivery notes to supplier invoices.
- Link delivery notes to supplier quotations.
- Maintain a complete procurement audit trail.

The system shall support supplier management by maintaining:

- Supplier profiles.
- Supplier contacts..
- Supplier compliance documents.
- Supplier quotation history.

Asset Management Requirements

The requirement-gathering exercise established that the Procurement Department is responsible for managing institutional assets.

The ERP system shall therefore include an Asset Management component

1. Asset Registration

When assets are acquired, the system shall:

- Assign a unique asset number.
- Record asset details.
- Record acquisition cost.
- Record supplier information.
- Record acquisition date.
- Record department ownership.
- Record asset location.

2. Asset Tracking

The system shall support:

- Asset movement tracking.
- Departmental asset allocation.
- Asset transfers.
- Asset maintenance history.
- Asset disposal records.

Management shall be able to determine the location and status of any institutional asset at any time.

3.Inventory Management

The Procurement Department also manages institutional inventory and stores.

Stock Monitoring

The system shall:

- Track current stock levels.
- Monitor minimum stock thresholds.
- Generate low-stock alerts.
- Monitor stock movement history.
- Support periodic stock audits.

This will help prevent stock shortages and improve inventory control

9. TICH SACCO

The TICH Savings and Credit Cooperative Society (SACCO) is an internal department established to provide financial services and support welfare initiatives for members of staff. In addition to its financial services, the SACCO oversees the management of the institution's cafeteria and selected hostel-related operations. The proposed ERP system shall integrate all SACCO operations into a centralized module to improve efficiency, transparency, accountability, and financial reporting

Membership Requirements

The requirement-gathering exercise established the following membership policies:

- Membership is open to all members of staff employed by TICH.
- Joining the SACCO is voluntary and is not a mandatory condition of employment.
- Every new member shall pay a one-time, non-refundable registration fee of **KES 500** before membership is activated.

- Upon successful registration, every member shall make a minimum monthly contribution of **KES 200** towards their savings account.
- Members may voluntarily contribute amounts higher than the minimum monthly contribution according to their financial capability.

Savings Management

The SACCO module shall maintain an individual savings account for every registered member. The system shall:

- Record all member contributions.
- Maintain an accurate running savings balance.
- Generate individual member account statements.
- Track contribution history.
- Notify members of overdue monthly contributions.
- Generate periodic savings reports for management.

Loan Management

The SACCO provides loan facilities to eligible members based on their accumulated savings.

The following business rules were identified during the requirements gathering exercise:

- Only registered SACCO members are eligible to apply for loans.
- Loan eligibility shall be determined based on the member's total accumulated savings.
- A member may borrow up to **three (3) times the value of his or her total savings**, subject to SACCO approval.
- Every loan application shall follow the SACCO's approval workflow before disbursement.
- The system shall automatically calculate the member's maximum loan eligibility based on current savings.
- The system shall maintain complete loan records, including application details, approval status, disbursement history, repayment schedules, outstanding balances, and loan clearance.

B. Front-End User's view

1. Home Page / Landing Page Portal

The public entry point acts as the primary tool to showcase TICH's transition into a registered Technical and Vocational College (TVC) and its strategic push across decentralized sub-counties.

- **A. High-Impact Carousel Slides:**
 - Dynamic, auto-playing media banners highlighting core institutional pillars.
 - *Unlimited slides, can be arranged*
- **B. Our Programs Snippet Matrix:**
 - A grid interface featuring "pinned" priority programs such as Community Health Providers (CHPs), Community Health & Development (CHDs), and Health & Development Technicians.
 - Each snippet displays the course duration, eligibility tier (e.g., NITA/CDACC artisan tracks), requirements, fee structure and a direct button to "Apply Now."
 - Includes a persistent anchor button: **"View All Academic Programs"** linking to the full academic catalog.
- **C. Our Research Hub Snippet:**
 - An editorial-style block emphasizing TICH's participatory community health research framework.
 - Features a prominent display of active field studies with a summary body describing ongoing interventions in local micro-economies.
 - Interactive CTAs: **"Read More"** (opens the specific project write-up) and **"View All Research"** (redirects to the primary Research Portal).
- **D. Integrated Events & Conferences Registry:**
 - A live feed pulling directly from the internal institutional calendar database.
 - Displays upcoming events, national healthcare symposiums, open days, and outreach schedules.
- **E. Latest Blogs Feed:**
 - Pulls the three most recent analytical pieces written by faculty, field researchers, or community coordinators.
 - Features social share shortcuts and estimation trackers for reading time.

2. About Us Module

This module details TICH's historical shift from a localized health program into an accredited, decentralized vocational training institution.

- A. About TICH: A comprehensive narrative outlining the college's operational philosophy—bringing structured, accredited training directly to the grassroots level rather than pulling rural learners to central urban campuses.
- B. Mission Statement: Dedicated text block focusing on empowering communities through accessible, competency-based education and sustainable local development models.
- C. Vision Statement: Dedicated text block focused on reversing the rural-to-urban "brain drain" of skills by anchoring youth and women within their local economies.
- **D. Institutional History Timeline:**
 - An interactive visual timeline tracking TICH's evolution from the **Saradidi Rural Health Program (1979)** and **Community Initiatives Support Services (CISS)** legacy up to its modern status as a recognized TVC.
- **E. Careers & Job Board:**
 - Directly integrated with the internal **HR Recruitment & Onboarding Module**.
 - Publicly renders open vacancies for tutors, sub-county coordinators, and administrative positions.
 - Features an online ingestion form allowing applicants to upload their CVs, certificates, and professional operational licenses directly into the HR screening pool.

3. Research Portal

This module serves as a repository for TICH's community-driven research and a bridge for securing external institutional collaborations.

- **A. Research Overview:** A conceptual description detailing TICH's evidence-based research methodologies that actively influence Kenyan health policy and grassroots micro-enterprise design.
- **B. Research Projects Directory:**
 - A categorized dashboard split into two distinct filter views: **Ongoing Projects** and **Completed Projects**.
 - Users can select individual project profiles to read abstract definitions, key field outcomes, and implementation timelines.
- **C. Defined Research Areas:** Focus tabs highlighting core institutional competencies, such as Community-Based Health Care (CBHC) legacy evaluation , digital health integration logs (e.g., m-Fundi operational efficiency tracking), and vocational training socio-economic impact studies.
- **D. Research Resources Center:** A secure, downloadable digital library containing policy briefs, peer-reviewed publications, open-access field data, and technical training toolkits.

- **E. Institutional Contact Details:** Direct contact lines, physical office mapping at the central Kisumu campus, and dedicated communication channels for academic inquiries.
- **F. Interactive Partnership Request Engine:**
 - A custom intake form for international NGOs, county governments, and academic institutions.
 - Captures organization type, proposed funding or collaboration scope, and target sub-counties. Submitting automatically alerts the Academic Board and the central Registrar.
- **G. Our Partners & Logos Marquee:** A clean logo slider displaying collaborating entities, including regulatory bodies (CDACC, NITA), funding foundations, and county public health departments.

4. Academics Module

This section serves as the digital course catalog, outlining TICH's specific training targets and vocational curriculum offerings.

- **A. Educational Framework Explanation:** Deep structural breakdown of the **Competency-Based Education and Training (CBET)** pedagogy. It highlights how learners receive formally recognized, certifiable credentials matching NITA and CDACC frameworks without requiring traditional university entry grades.
- **B. Departmental Breakdown & Curricula:**
 - Detailed program blueprints for the **Community Health and Development (CHD)** department and **Vocational Training** sectors.
 - Specific syllabus parameters, modules, practical laboratory credit requirements, and credit transfer evaluation rules.

5. Admissions Module

This portal handles student ingestion, accommodating both tech-enabled online applicants and non-tech or rural learners.

- **A. Digital Admission Form:**
 - A multi-step application wizard linked to the central Academic Registrar's database workspace.
 - Captures bio-data, national identity details, and high-school or practical work backgrounds for **Prior Learning Assessments (RPL)** uploads.
 - Dropdown selector for the applicant's closest administrative **Sub-County Hub** to ensure localized placement.
- **B. Manual Application Guide:**

- A step-by-step instructional guide and downloadable paper form for applicants with limited internet connectivity.
- Directs applicants on how to submit physical forms via field-level mobile ingestion agents or local Sub-County Coordinator offices.
- Details acceptable local bursary and financing pathways, such as Constituency Development Funds (CDF) or Ward allocations.

6. News & Events Hub

- **A. Events Calendar:** A filterable calendar detailing graduation ceremony, community health drives, exam registration schedules, and module enrollment cycles.
- **B. Conferences Dashboard:** Highlights major research or vocational education conferences hosted or attended by TICH, including abstract submission portals and registration fee parameters.
- **C. Interactive Gallery:** A media archive organized by folders (e.g., Practical Skills Assessments , Sub-County Training Hub Launches, and Active Work-Study Project Sites). Images to have captions.
- **D. Institutional Blog:** An publishing platform where instructors post educational notes and community leaders write about grassroots economic empowerment.

7. TICH SACCO Portal

This portal functions as the primary digital interface for the institution's integrated micro-financing and economic safety net.

- **Welcome Note & Vision:** Introduction to the SACCO's role in driving economic self-reliance for graduates, tutors, and affiliated community health workers.
- **Services Offered Portfolio:**
 - Detailed breakdowns of savings options, dividend options, and asset financing programs.
 - Specifically highlights Toolkit Loans designed to help community college graduates buy tools or equipment (e.g., home care medical kits, solar installation gear) to launch micro-enterprises.
- **Purpose & Strategic Mandate:** Explains how the financial entity anchors local youth and women within their micro-economies, helping them build small businesses without needing traditional commercial bank history.
- **How to Join & Membership Qualifications:** Lists the clear entry criteria for TICH staff , active students within work-study systems , and local artisan mentors.
- **Membership Matrix:** Outlines member responsibilities, minimum monthly share capital contributions, voting rights, and guarantor obligations.

- **Application Process Workflow:** An online application portal that captures registration details and paths them to the internal **TICH SACCO / Enterprise Manager Interface** for review, credit scoring, and activation.
- **Contact Panel & Communications Hub:** Features a dedicated contact form that sends messages straight to the SACCO credit department, along with its official, direct email address eg: sacco@tichinafrica.org.

8. Support Us (TICH Fund)

- **About the TICH Fund:** An informative narrative detailing how contributions are utilized. Funding directly backs student bursaries , helps construct decentralized community college facilities , and secures tools for student work-study projects.
- **Donation Details & Payment Gateways:**
 - Secure endpoints for financial transactions.
 - Localized integration for **M-Pesa Paybill** numbers (with automated account validation scripts linked to the finance ledger).
 - Standardized international bank wire transfer numbers, SWIFT routing details, and tax-exempt documentation download links for corporate sponsors.

9. Student Portal Gateway

This page serves as a secure landing gate for student authentication. Once logged in, it connects users to a comprehensive personal dashboard:

Student Portal Sub-Modules	Core Functional Capability
Academic Records	View enrolled courses, download lesson resources , and track cumulative assessment marks.
Attendance Monitor	View geo-fenced field logs and check if attendance drops near the 90% exam threshold.
Work-Study Ledger	Track logged hours on community field assignments and monitor tuition offset balances.

Financial Services	View tuition statements, apply for SACCO toolkit loans, or track external CDF/Ward bursaries.
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10. Tutor Portal Gateway

A secure authentication node for faculty, visiting lecturers, and community-level artisan mentors. It grants immediate access to day-to-day teaching tools:

- A. Lesson Plan Manager: Interface to build, save, and route upcoming weekly lesson outlines to the respective HOD or Academic Registrar for verification.
- B. Digital Attendance Register: Enables tutors to open geo-fenced registers matching their specific sub-county hub and log daily class participation.
- C. CBET Competency Grading Hub: A spreadsheet interface used to score written CATs, field practicals, and clinical or technical skills lab assessments.
- D. Exam Lifecycle Hub: Secure portal used to submit draft exam templates to the Academic Board and Registrar for moderation and supplementary exam tracking.

11. Careers

This module renders active vacancies and hosts the applicant data-ingestion portal. It is configured to display standard institutional marketing blocks dynamically based on the listing category.

A. Dynamic Promotional Content Blocks

When a user selects or filters for "Internship" opportunities, the frontend automatically pulls and displays the following certified institutional text:

Begin Your Journey With Us

An internship at TICH is more than a placement—it is an opportunity to contribute to community transformation while building the foundation for your professional future. If you are passionate about community health, development, research, or organizational support, we welcome your application.

Join Us in Advancing Community Health & Development

The Tropical Institute of Community Health and Development (TICH) offers a dynamic Internship Program designed to equip emerging professionals with hands-on experience in community health, development practice, research, and organizational operations. Our internships provide meaningful learning opportunities while supporting the Institute's mission to strengthen community health systems and drive sustainable development.

Interns at TICH work on real projects, contribute to impactful initiatives, and gain exposure to the day-to-day operations of a leading community health institution. All assignments are supervised by

experienced staff who provide mentorship, guidance, and structured feedback throughout the internship.

Why Intern at TICH?

- **Practical learning experience** aligned with your academic background and career interests.
- **Exposure** to community health and development programs.
- **Mentorship** from seasoned professionals.
- **Opportunities** to contribute to meaningful projects.
- **A supportive, inclusive, and diverse** learning environment.

B. Public Application Ingestion Form

Every active listing renders a secure multi-part form requiring the applicant to provide the following data arrays before submission is unlocked:

- **Personal Details:** Full Name, Email Address, Phone Number, and physical/postal coordinates.
- **Academic/Professional Context:** Highest Level of Education, Current Institution/Organization, and Area of Specialization.
- **File Upload Vault:**
 - *Curriculum Vitae (CV):* Mandatory upload (PDF format, capped at 5MB).
 - *Cover Letter:* Optional upload or plain text area.
 - *Certificates/Transcripts:* Multi-file attachment slot.

12. Footer Module

A persistent component rendered across all public webpages and portal entry screens, organizing essential institutional links and compliance details.

- **A. Newsletter Subscription Option:** An entry field for email capture that feeds directly into the communications registry. This lets the college push notifications regarding upcoming admission windows, research findings, and community college milestones.
- **B. Logo & Copyright Information:** Displays official institutional branding and organizational credentials:
© 2026 TICH Enterprise Resource Platform. Registered Technical and Vocational College (TVC), Kenya. All Rights Reserved.
- **C. Primary Site Navigation Array:** Column arrangements pointing directly to major sections: Home, About Us, Research Portal, Academics Catalog, and Admissions Guide.

- D. Operational Quick Links: Shortcuts to high-traffic portal entry nodes: Student Portal, Tutor Portal, Staff Portal ESS , TICH SACCO Account Login, and open Career Vacancies.
- E. Contact Us Information Array:
 - i. Physical Address: Full corporate tracking location for the Central Campus in Kisumu, Kenya, complete with an embedded map link and an index of active sub-county training hubs.
 - ii. **Email Coordinates:** General inquiries: info@tichinafrica.org; Academic admissions tracking: admissions@tichinafrica.org.
 - iii. **Phone Channels:** Central campus registrar helpdesk numbers and dedicated student support hotlines.
 - iv. **Social Channels Array:** Connected social media link icons linking directly to TICH's active profiles on X (Twitter), LinkedIn, Facebook, and YouTube for field video storage.

C. Platform Users

1. Institutional Governance & High-Level Executive Users

These accounts govern macro-level institutional configurations, corporate policy execution, and final digital authorization workflows.

- **Chief Executive Officer (CEO):** The ultimate global system administrator; holds decisive digital sign-off authority required to activate newly provisioned academic departments, courses, and major financial configurations.
- **IDM (Institutional Development Management):** Executive-level workspace managers tracking high-level administrative parameters.
- **Academic Board Members:** Institutional policy-engine users responsible for establishing grading frameworks, passing criteria, and completing summative exam paper moderation.

2. Academic Leadership & Core Administrative Staff

Back-office users executing student lifecycle operations, master data cataloging, and testing verification sequences.

- **Academic Registrar:** Leads the central back-office student terminal, configuring the dynamic trimester calendar, running credit transfer equations, managing waitlists, and overseeing transcript issuance.
- **Heads of Departments (HODs):** Department-level administrators balancing faculty workloads, designing course units/modules, routing syllabus outlines for approval, and executing departmental self-audits.

- *Covers specific HOD profiles across: Community Health and Development (CHD), Nursing, Vocational Training, Information Communication Technology (ICT) etc.*
- **Examination Officer:** Manages the secure script injection flow and coordinates exam marking operations.
- **Central Admissions Desk Officers / Admin Reps:** Front-office personnel who process physical or digital intake packages, verify applicant documentation, and generate permanent registration numbers.

3. Operational, Audit & Support Staff Profiles

Departmental backend accounts that maintain compliance, track assets, oversee human capital, and manage institutional workflows.

- **Human Resource (HR) Manager / HR Staff:** Governs the entire employee lifecycle, manages staff qualifications, tracks professional development credits, triggers contract renewal sequences, and oversees the disciplinary record hub.
- **Chief Finance / Finance Officer:** Authorizes net salary remittance engines, configures statutory tax bands (PAYE, NSSF, SHA, AHL), tracks accounts receivable/payable, balances donor accounts, and confirms supplementary exam fees.
- **Internal Auditor:** Independent dashboard profile tracking complete accounting pathways and verifying transaction histories against central ledger controls.
- **Monitoring & Evaluation (M&E) Officer:** Accesses unalterable, system-wide historical audit trails and log data to generate analytic reports detailing long-term regional performance parameters.
- **Quality Assurance (QA) Officer:** Controls the central QA command center; deploys custom, multi-department digital assessment sheets, checks lesson plan compliance, and issues Quality Corrective Action flags.
- **Procurement Team / Officers:** Vets suppliers, captures invoices, executes three-way match validations against physical delivery parameters, and passes cleared files to Finance.
- **Infrastructure / Asset Management Staff:** Tracks institutional asset status and resources across the physical campus properties.
- **Research Officers / Field Researchers:** Compiles reviewed data, manages active field studies, and publishes policy briefs inside the public-facing repository.
- **ICT Systems Administrator / Core Technical Team:** Maintains system, manages role-based access control configurations, handles database partitions, assist users to operate within the platform and steps in directly to resolve escalated password reset locks.
- **Communications & Marketing Staff:** Holds specific editing rights to style public job openings, tracks referral telemetry, and manages outreach communications.

4. Instructional, Teaching & Field Operations Staff

The front-line personnel bridging central campus operations with decentralized sub-county training activities.

- **Tutors / Lecturers / Faculty:** Instructors who submit digital lesson plans, create content hubs, log grades, and record classroom attendance.
- **Visiting Lecturers:** Temporary faculty members given transient access to specific course-delivery models and grading grids.
- **Health & Nursing Tutors:** Faculty profiles carrying strict, system-monitored professional operational licenses that are verified before unit allocation.
- **Affiliated Local Artisans / Trade Mentors:** Field-level practical industry experts certified under NITA/CDACC regulations to guide and log competency checklists within remote sub-county hubs.
- **Invigilators (Exam Supervisors):** Proctors assigned to active exam session spaces who collect completed scripts and log testing logs into the platform.
- **Independent Internal / External Examiners:** Independent subject-matter evaluators assigned to scripts that are flagged for remarking.
- **Community Mobilizers / Field Intake Agents:** Local field workers who enroll rural learners during field outreach programs.

5. Enterprise, Sacco & Auxiliary Service Users

Users specialized in running commercial, welfare, and physical resource facilities connected to the broader TICH ecosystem.

- **TICH SACCO & Enterprise Manager:** Manages staff cooperative databases, tracks share contributions, reviews loan scores, and processes graduate toolkit loans.
- **SACCO Credit Department / Loan Review Officers:** Standard SACCO backend staff who monitor outstanding balances and track loan payment cycles.
- **Hostel Accommodation Administrator:** Coordinates institutional housing bookings, checks real-time room counts, and tracks monthly student rent entries.
- **Cafeteria Operations Staff:** Manages sales logs, checks internal staff memberships, and reconciles transactions through an isolated accounting bank structure.

6. End-Users, Self-Service Tiers & External Stakeholders

Public, student, and third-party portal profiles that interact with specialized front-end interfaces and integration nodes.

- **Regular Students:** End-users logging into self-service portals to access timetables, trace attendance warnings, check published CAT grades, track tuition ledgers, and query the 24/7 AI chatbot.
- **Applicants / Prospective Students:** External users accessing public forms to upload bio-data, secondary school certificates, and transaction codes to initiate admissions tracking.
- **Job Applicants / Tutor Candidates:** Job seekers utilizing the career board interface to upload CVs and professional licenses into the HR ingestion pool.
- **Alumni:** Former graduates possessing lifetime entry privileges to the portal vault to safely fetch secure digital transcripts and authenticated completion credentials.
- **Sponsors / Institutional Donors:** Third-party organizations or grant providers utilizing specialized billing modules to trace funds and receive foreign disbursement reports.
- **Verified Suppliers / Vendors:** External merchant profiles provisioned post-procurement to track independent billing balances, submit raw invoices, and view settlement statements.
- **Guests / Public Web Visitors:** General public users accessing unrestricted content blocks, reviewing open blogs, or subscribing to the institutional newsletter.

D. Engagement

To establish a resilient, modern infrastructure for the TICH Enterprise Resource Platform (ERP), the user engagement landscape must be architected across three distinct deployment environments: Web, Mobile, and Desktop. This separation maximizes interface usability, and optimizes backend data security.

1. Web Application Portal Ecosystem (Universal Access Layer)

The web application serves as the primary, universal engagement tier accessible via standard web browsers. It hosts cross-departmental dashboards, handles student self-service gateways, and manages secure, public-facing ingestion channels.

Administration & Executive Layer (CEO):

- Accesses the administrative interface to monitor multi-tiered, cross-departmental institutional planning matrices.
- Executes dual-authorization workflows
- Reviews high-level Quality Level Reports compiled from unalterable system logs.

Academics & Registrar Console:

- Houses the complete Academic Board Portal for processing policy definitions, pass marks, and marking guidelines.
- Allows the Academic Registrar to map out dynamic trimester structures, configure enrollment windows, and review waitlists or deferrals.

Finance Department Hub:

- Operates accounts receivable and accounts payable sub-modules, handles donor-funded project tracking, and monitors real-time payment settlements.
- Oversees automated net salary remittance and manages the localized statutory tax engine.

Human Resource Management Portal:

- Runs full employee lifecycle pipelines, configures employment contract categories, and tracks dynamic contract end-dates.
- Manages the secure incident registry and disciplinary record dashboard.

Tutors / Instructional Space:

- Utilizes single-sign-on (SSO) integrations and accesses learning materials
- Accesses the continuous assessment terminal to manage grading grids and log competency checklists.

Students Portal Interface:

- Logs into web-based self-service spaces to complete unit registration, view personalized timetables, and trace attendance milestones.
- Accesses real-time tuition statements, pulls downloadable payment receipts, and triggers official transcript requests.

Guests / External Stakeholders (Restricted Access):

- Interacts with public entry points to browse prioritized programs or read the participatory community health research hub and general information about TICH.
- Completes multi-step digital forms to submit applicant information, upload certificate files, or subscribe to the newsletter registry.
- Renders open career positions on the public job board to allow applicants to route CVs directly into the screening pool.

2. Mobile Application Ecosystem

The mobile application is optimized for lightweight execution, touch-based interactivity, hardware camera utilization, and reliable operation

- **Tutors & Local Artisan Mentors:**
 - Can create and submit lesson plans
 - Activates the mobile camera module to capture and upload photos of physical signed attendance logs as unalterable validation files.
 - Logs practical trade milestones on decentralized competency checklists right from local training sites.
- **Students Self-Service Portal:**
 - Receives immediate push notifications regarding upcoming assignment deadlines, fee balances, or sudden timetable shifts.
 - Engages with the built-in, 24/7 multi-lingual AI Chatbot Assistant to query admission rules or any other relevant information that they may need.
 - View their fees, examination results, download exam cards etc.
- **Staff Portal & Community Field Mobilizers:**
 - Employs low-bandwidth, offline-ready mobile web interfaces to input applicant profiles on-site during county entry drives.
 - Caches collected biographical data and certificate credentials locally on the device, triggering an automated synchronization with the central database once an internet connection is established.
 - Submits on-the-go leave applications and monitors individual leave balances or approval flows.

3. Native Desktop Application Terminals (Back-Office Heavy Operations)

The desktop application environment is built specifically for data-intensive administrative tasks, large-scale system setups, and local network resource management within campus facilities.

- **Administration & Operations Workspace:**
 - Handles heavy multi-tiered institutional budget configurations, consolidates cross-departmental allocations, and enforces strict prior-month submission deadlines.
 - Manages centralized repositories for compliance files, registration documents, and regulatory alignment records.
- **Academics & HODs Workspace:**
 - Coordinates the multi-user Timetable Builder Workspace to construct classes/lessons, master schedules, and exam room matrices.

- Evaluates non-standard entry paths and runs calculation scripts within the Recognition of Prior Learning (RPL) engine.
- Provisions new departments, builds core prerequisite logic, and structures clinical block systems for nursing tracks.
- **Finance & Accounting Terminal:**
 - Compiles monthly electronic bank transfer files (EFTs) and processes complex multi-currency conversion routines for donor disbursements.
 - Synchronizes real-time ledger records and transaction trails directly with QuickBooks to ensure clean bank reconciliations.
 - Implements automated compilation workflows for annual cumulative P9 tax documents.
- **Human Resource Suite:**
 - Coordinates staff records to manage qualifications, professional development credits, and valid professional licenses.
 - Controls granular role-based access permissions, manages user account safety bounds, and monitors unalterable global activity audit logs.

E. Enhancements

Biometric & Security

- Multi-layered security architecture:
 - **Two-Factor Authentication (2FA):** Mandatory for all staff and optional for students (via SMS, WhatsApp, Email, or Authenticator App).
 - **Role-Based Access Control (RBAC):** Granular permissions (e.g., a lecturer can only see their courses and students, while an Academic Registrar sees all departments).
 - **Audit Trails:** Every action (grade change, fee waiver, data export) is logged with user, timestamp, and reason.
 - **Data Encryption:** End-to-end encryption for sensitive data (medical records, financial info).
 - Compliance with **Kenya Data Protection Act (DPA)**, GDPR principles, and ISO 27001 standards.
 - Session timeout, IP whitelisting for admin access, and AI-based anomaly detection (e.g., unusual login locations).
 - Regular automated security audits and vulnerability scanning.

Digital Credentials

- Secure issuance of transcripts.
- Tamper-proof design with embedded metadata (graduate name, program, classification, graduation date).
- Bulk generation for graduation ceremonies.
- Downloadable in PDF and shareable formats (LinkedIn, email).
- Lifetime access for alumni through the alumni portal.
- Revocation capability for cases of academic misconduct.

AI Chatbot for Student Support

- Intelligent virtual assistant available 24/7 on the mobile app and website.
- **Key Capabilities:**
 - Answer admissions queries (entry requirements, application status, deadlines).
 - Provide real-time fee balance and payment options.
 - Academic advising: suggest courses, check graduation progress, prerequisite warnings.
 - Timetable and venue information.
 - Result and transcript requests.
 - General campus information (holidays, events, policies).
- Escalation to human support (live chat with admissions/registrar team) when complex issues arise.
- Multi-language support (English + Kiswahili).
- Continuous learning: improves answers based on frequent queries and feedback.

F. Data Management Framework

Overview

The proposed TICH ERP Platform will serve as the institution's centralized repository for academic, administrative, financial, and operational information. Data will be collected, processed, stored, and secured in accordance with the Kenya Data Protection Act, institutional policies, and applicable regulatory requirements.

The platform will support the collection and management of data from students, staff, administrators, alumni, applicants, sponsors, and external stakeholders.

Student Data

The system will maintain comprehensive student records throughout the student lifecycle, from application and admission through graduation and alumni engagement.

Category	Data Collected
Personal Information	Full Name, Passport Photo, Gender, Date of Birth, Age, National ID Number, Passport Number (if not Kenyan), Nationality, Religion (optional)
Contact Information	Mobile Number, Alternative Phone Number, Email Address, Postal Address, Physical Address, Emergency Contacts
Next of Kin / Guardian Information	Parent/Guardian Names, Relationship, Phone Numbers, Email Addresses, Physical Address
Admission Information	Application Number, Admission Number, Program Applied For, Entry Qualifications, Admission Status, Admission Date
Academic Records	KCSE/KCPE Results, Previous Institution Records, Certificates, Diplomas, Transcripts, Academic Progress Reports, Examination Results
Registration Information	Registered Units, Courses, Semester Registrations, Academic Status
Attendance Information	Attendance Logs, Attendance Percentages, Class Participation Records
Financial Information	Fee Invoices, Payments, Balances, HELB Details, Scholarships, Sponsorship Information
Medical Information	Medical Conditions, Allergies, Special Needs, Disability Status (where applicable)
Accommodation Information	Hostel Allocation, Room Details
Disciplinary Records	Warnings, Cases, Disciplinary Decisions, Appeals
Graduation Information	Graduation Status, Classification, Certificates, Transcripts
Alumni Information	Employment Status, Alumni Membership, Professional Information, Tracer Study Data

Staff Data

The platform will maintain employee records to support administration, payroll, performance management, and institutional reporting.

Category	Data Collected
Personal Information	Full Name, Gender, Date of Birth, National ID Number, Passport Number (If not Kenyan), Nationality
Contact Information	Phone Number, Email Address, Physical Address, Emergency Contacts
Employment Information	Employee Number, Department, Position, Job Grade, Contract Type, Employment Date
Professional Qualifications	Academic Certificates, Professional Certifications, Licenses, CV/Resume
Payroll Information	Salary Details, Bank Information, PAYE Details, NHIF Information, NSSF Information, Pension Information
Attendance Records	Attendance Logs, Leave Records, Overtime Records
Performance Information	Performance Reviews, KPIs, Appraisals, Training History
Academic Information	Courses Taught, Teaching Workload, Research Projects, Publications
Security Information	Login History, User Access Rights, Audit Logs

Administrative and Institutional Data

The platform will support the collection and management of institutional data required for governance, operations, compliance, and reporting.

Category	Data Collected
Academic Structure	Departments, Programs/Courses, Units, Curriculums
Examination Data	Examination Schedules, Assessment Records, Results, Moderation Reports, Examination timetables
Financial Data	Fee Structures, Budgets, Invoices, Receipts, Procurement Records, Financial Statements
Human Resource Data	Organizational Structure, Staffing Levels, Recruitment Records, Performance Reports

Communication Data	Announcements, Notifications, SMS Logs, Email Logs, and Communication Records
Compliance Data	Accreditation Records, Regulatory Reports, Audit Reports, Quality Assurance Documents
Security Data	User Accounts, Login Records, Activity Logs, Access Permissions, System Audit Trails
Analytics Data	Student Performance Metrics, Enrollment Statistics, Retention Rates, Financial Indicators, Institutional KPIs

Conclusion: Driving Institutional Excellence

The implementation of the TICH Enterprise Resource Platform (ERP) represents a fundamental step in modernizing TICH's operational infrastructure. By consolidating fragmented manual processes into a unified, secure, and data-driven ecosystem, TICH will not only achieve immediate improvements in operational efficiency and accountability but will also establish a robust digital foundation for long-term institutional growth.

This platform is specifically engineered to align with TICH's unique model, ensuring that every departmental function is standardized and transparent. By empowering leadership with real-time analytics, streamlining student onboarding and academic administration, and automating complex financial and HR workflows, this project will significantly enhance service delivery for students, staff, and stakeholders alike.

Ultimately, this ERP solution is a technological upgrade and an institutional transformation program that reinforces TICH's commitment to quality education, community-based research, and sustainable development. We are fully prepared to deliver this scalable, compliant, and user-centric platform, ensuring the institution remains at the forefront of higher education and vocational training in Africa.